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Review

Machine learning is rewiring Li-ion battery RUL prediction: A systematic review and outlook

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ABSTRACT

Accurate prediction of the remaining useful life (RUL) of lithium-ion batteries is pivotal for the safe and economical operation of electric vehicles and stationary energy storage systems. However, the rapid expansion of machine-learning and hybrid prognostic models has fragmented evaluation practices and weakened the reliability of cross-study comparisons. This work provides a structured review of data-driven RUL prediction through a unified taxonomy linking each model family to its learning target, inductive assumptions, data requirements, and common failure modes. Methodologically, this review conducts a systematic literature mapping and fine-grained bibliographic tagging of 996 publications, with emphasis on the 2017–2025 period, and quantitatively analyzes research trends, dataset usage, validation protocols, and reporting conventions to clarify where performance claims are comparable. This work further summarizes the outstanding challenges at the data, methodological, and application levels, including dataset limitations and inconsistent end-of-life definitions, generalization and benchmarking fragmentation with insufficient uncertainty calibration, and deployment, safety, and standardization constraints.

1. Introduction

Rapid electrification in transportation and pervasive mobile electronics are driving steep growth in battery demand, with Li-ion remaining the dominant technology due to its high specific energy and long service life [1–3]. Despite these advantages, external stressors such as elevated temperature, irregular and high-rate charging, and prolonged use accelerate degradation via well-studied mechanisms including loss of lithium inventory through solid electrolyte interphase/cathode electrolyte interphase (SEI/CEI) growth, lithium plating, particle cracking, and transition-metal dissolution and cross-talk [2,4–7]. In this review, remaining useful life (RUL) refers to the remaining first-life cycles from the current state until a predefined end-of-life (EOL) threshold, which for Li-ion traction and stationary applications is commonly taken as 70%–80% of initial capacity [8,9]. Reliable RUL estimates enable safer operation and predictive maintenance by reducing unplanned downtime and maintenance costs [10], improving asset availability, and supporting both warranty management and fleet-level risk control [11,12]. In the context of circular-economy pathways, accurate RUL estimates at both the cell and pack levels can guide triage decisions between second-life reuse and direct recycling, thereby enhancing techno-economic viability and reducing the environmental burden associated with premature disposal or inefficient end-of-life management [9,13]. Finally, accurate RUL modeling feeds back into battery design and accelerated testing by quantifying how early-cycle signatures relate to lifetime, shortening iteration loops in materials and protocol development [11,14].

Across the prognostics and health management (PHM) literature, RUL prediction methods are typically classified into three categories: model-based approaches, data-driven approaches, and hybrid approaches [11,15]. Model-based methods rely on physics or stochastic-degradation models to forecast future evolution, offering explicit mechanistic interpretability and principled extrapolation when their assumptions hold [7,11,16]. However, such methods often transfer poorly across duty cycles and operating environments, and their deployment can be hindered by parameter identifiability and computational burden [16]. Against

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this backdrop, data-driven approaches to RUL prediction have emerged as a natural development. They learn mappings directly from historical measurements to RUL; once trained, they are relatively straightforward to deploy and can generalize across moderate variations in operating conditions [8,15]. Representative data-driven approaches for Li-ion battery RUL prediction span both classical and deep learning methods. Classical algorithms such as support vector regression (SVR) and tree-based ensembles (e.g., random forests (RFs) and gradient boosting) provide strong baselines on tabular features, offering solid accuracy with relatively modest data requirements and limited feature engineering [17]. On the deep learning side, long short-term memory (LSTM) networks and related gated recurrent architectures explicitly capture temporal dependencies in degradation trajectories and have therefore become canonical choices for sequence-based RUL estimation [18]. One-dimensional convolutional neural networks (1D-CNN) further exploit local patterns in current-voltage-temperature time series or time-frequency representations [19–21], automatically extracting salient degradation features with compact architectures. More recently, temporal convolutional networks (TCNs) and Transformer-based models have leveraged dilated convolutions and self-attention [22,23], respectively, to model long-range dependencies and interactions across measurement channels such as voltage, current, and temperature, and are emerging as promising candidates for RUL prediction under complex, nonstationary operating profiles.

However, the strengths of these data-driven models come at a cost: they are often opaque, rely on large amounts of well-curated degradation data, and may become brittle when operating conditions deviate substantially from those seen during training. These limitations have motivated growing interest in hybrid and physics-guided approaches that embed domain knowledge or mechanistic constraints into the learning process in order to improve robustness, data efficiency [24], and interpretability [16,25–27]. In the RUL context, hybrid methods typically couple data-driven learners with electrochemical or equivalent-circuit models in one of several ways [28,29]. A common pattern is to use mechanistic models to generate physics-informed features or intermediate states, which are then fed into machine-learning regressors for RUL estimation [25]. Alternatively, learning architectures can be regularized by physical constraints or residual structures, for example by training neural networks to model the discrepancy between simplified physics and observed degradation, or by enforcing monotonicity and operational limits in the loss function [30]. While these approaches currently represent a smaller fraction of the literature than purely data-driven models, they offer a promising route toward combining the flexibility of machine learning with the structure and extrapolation capabilities of physics-based modeling [31–33].

This review focuses on machine learning-based RUL prediction for Li-ion batteries and does not attempt a full treatment of model-based methods. Within this scope, we summarize the recent landscape of representative learning-based approaches, including their basic principles, application scenarios, strengths, and limitations, together with an outlook on future research. Unlike broader surveys organized mainly by method categories or application domains, this review adopts a more explicit analytical perspective by linking public datasets, input representations, model families, validation settings, and the conditions under which reported performance claims should be interpreted.

This review goes beyond a purely narrative overview by incorporating a quantitative analysis of the literature. Specifically, we perform systematic bibliographic statistics on method usage and temporal trends, and many of our conclusions are supported by these literature-level results. In our discussion of future directions, we likewise ground our arguments in the quantitative patterns identified from this literature analysis and briefly discuss privacy-preserving and collaborative learning as one promising future direction under restricted data-sharing settings, with federated learning (FL) as a representative framework.

Overall, the main contributions of this review can be summarized as follows.

• *Synthesis of public datasets and benchmark relevance.* We provide an updated review of public battery aging datasets and discuss not only their experimental characteristics, but also their methodological relevance to RUL prediction. In particular, we highlight how dataset design, end-of-life definitions, protocol heterogeneity, and diagnostic richness influence benchmark utility, model selection, and the interpretation of reported results.

• *Systematic and structured review of data-driven methods.* We systematically cover a broad range of data-driven RUL prediction techniques, with emphasis on their modeling principles, typical input representations, data requirements, application scenarios, and principal limitations. Rather than reviewing algorithms in isolation, we highlight how different method families align with different prognostic settings.

• *Quantitative and comparability-oriented literature analysis.* We complement the qualitative synthesis with bibliographic statistics on method usage and temporal trends, and further examine dataset usage, validation protocols, reporting conventions, and end-of-life definitions. In this way, the review emphasizes not only what results have been reported, but also under what conditions those results are meaningfully comparable.

• *Discussion of privacy-preserving and collaborative learning.* We discuss privacy preserving learning, including federated learning, as a promising but scenario-dependent future direction for battery RUL modeling under restricted data-sharing settings.

Beyond these core contributions, this review also discusses the main limitations of existing datasets, summarizes commonly used data-driven RUL methods, and presents schematic illustrations of representative neural architectures.

The remainder of this article is organized as follows which is also shown in Fig. 1: Section 2 provides an overview of the field and describes how we collect and process the literature data used in this review. Section 3 introduces the popular datasets used in recent RUL prediction. Section 4 introduces the main machine learning methods for RUL prediction, outlining the general structure of each approach, the application scenarios in which it is particularly effective, and the corresponding limitations where appropriate. Section 5 discusses the key challenges and open problems that persist in current research. Section 6 discusses promising directions for future work. Finally, Section 7 concludes the paper.

2. Overview

This section provides an overview of the field and briefly summarizes the literature collection strategy used in this review.

Relevant studies were retrieved primarily from Web of Science and Google Scholar using search terms related to lithium-ion batteries, RUL, and machine-learning-based prediction. After screening, 996 publications were retained for the bibliographic analysis used throughout this review. Full search details are provided in the Supplementary Information [34–36]. In the following sections, a comprehensive statistical analysis of this research field will be conducted, focusing on aspects such as publication trends over time and the evolving emphasis of different research methodologies.

To contextualize the subsequent methodological review, we first quantify publication trends for model-based versus data-driven RUL studies over 2017–2025. Yearly records are collected for both categories and their annual shares are visualized to clarify

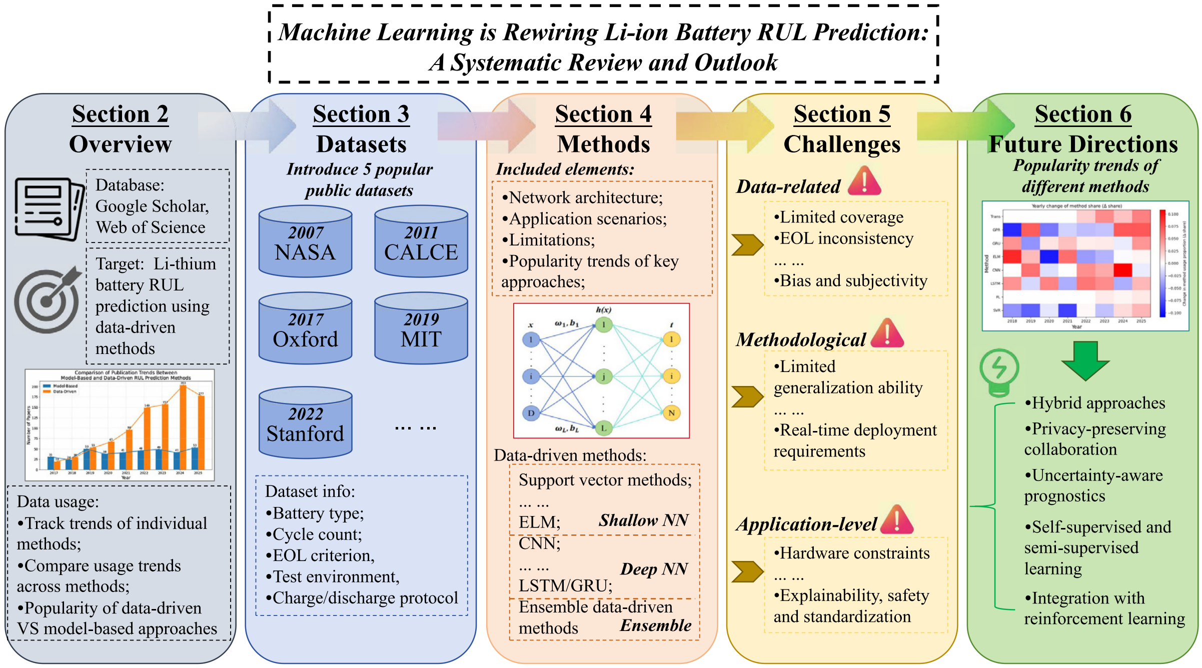

Fig. 1. Structure of this review.

the field's shift from model-based pipelines toward data-driven and hybrid prognostics.

As shown in Fig. 2, except for the year 2017, the number of data-driven RUL studies has consistently exceeded that of model-based research. Moreover, model-based studies have not exhibited a significant increase in quantity over time, whereas data-driven RUL research has experienced an explosive growth in recent years, far surpassing model-based approaches in volume. In recent years, more than 150 data-driven RUL prediction papers on lithium-ion batteries have been published annually, indicating that a focused investigation into machine learning-based RUL prediction is both necessary and timely [34,37,38].

Furthermore, this review summarizes the three most frequently used RUL prediction methods on a yearly basis. As shown in Fig. 3, earlier work predominantly relied on methods such as extreme learning machine (ELM) and SVR. In contrast, since 2022, LSTM- and CNN-based models have become the prevailing approaches, reflecting a clear methodological shift in the literature.

3. Public datasets for battery degradation and RUL studies

In this section, a concise overview is presented of the public datasets most commonly used for machine-learning-based prediction of RUL of lithium-ion batteries, which is also illustrated in a concise manner in Fig. 4. These datasets are discussed here not only as public resources for methodological development, but also as benchmarks whose design choices directly affect model selection, task formulation, and the interpretation of reported results. Particular attention is therefore paid to their experimental objectives, protocol characteristics, end-of-life definitions, and diagnos-

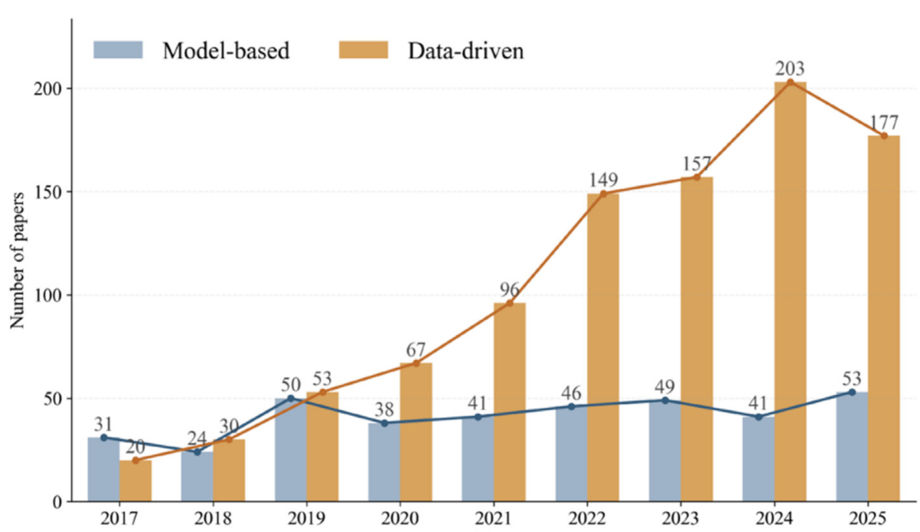

Fig. 2. Comparison of publication trends between model-based and data-driven RUL prediction methods.

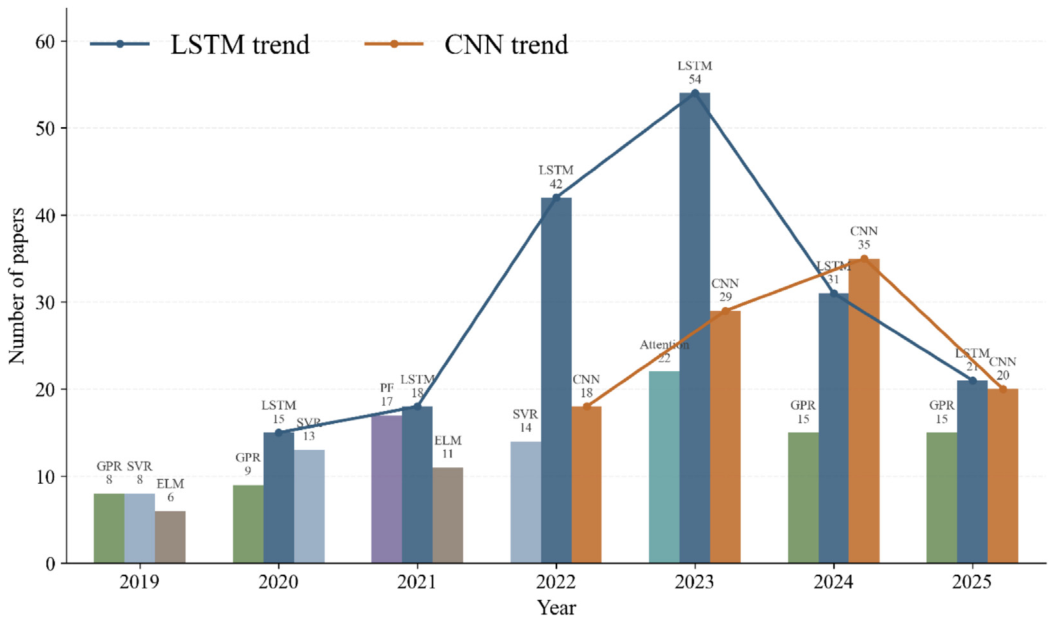

Fig. 3. Top-3 RUL prediction methods.

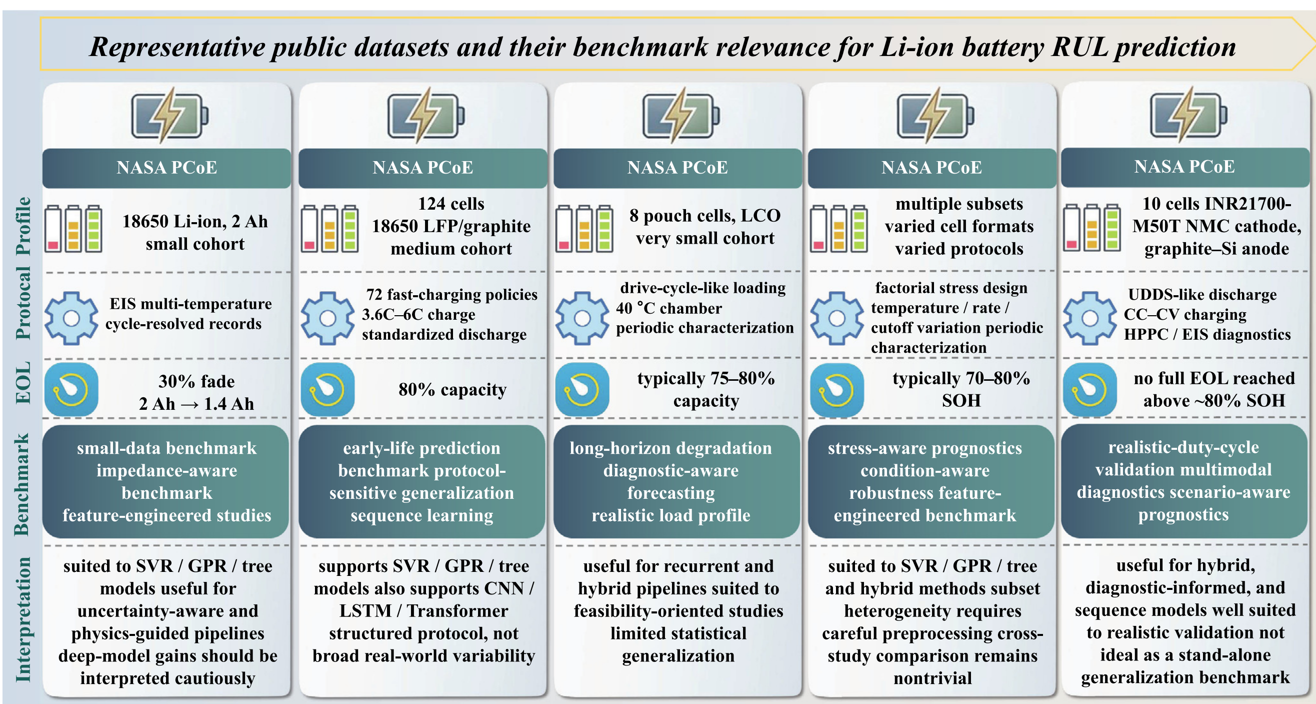

Fig. 4. Information of the popular public datasets.

tic richness, because these factors strongly influence both benchmark utility and the comparability of published RUL studies.

The datasets discussed in detail were selected as representative benchmarks rather than as an exhaustive list of all available battery aging resources. The selection was guided by five criteria: (i) public accessibility and sufficient documentation; (ii) frequent adoption or clear methodological influence in battery state of health (SOH) and RUL studies; (iii) availability of cycle-resolved capacity, voltage, current, temperature, or diagnostic measurements that support feature construction for machine learning (ML); (iv) clearly defined cycling protocols and capacity-fade or EOL information for constructing RUL-related targets; and (v) complementary benchmark roles across different ML scenarios, including small-data regression, early-life prediction, sequence learning, stress-aware generalization, realistic-duty-cycle validation, and diagnostic-informed modeling. Accordingly, NASA PCoE, MIT fast-charging, Oxford OBD-1, CALCE, and Stanford EV real-driving profiles are discussed in detail because they collectively cover the major dataset regimes encountered in ML-based RUL studies: impedance-aware small-data prognostics, early-life fast-charging prediction, drive-cycle-like sequence modeling, stress-factor-controlled aging analysis, and realistic-duty-cycle validation with multimodal diagnostics.

As shown in Fig. 5, Lithium-ion battery degradation arises from a combination of electrochemical side reactions and mechanical damage. Over repeated cycling, growth of the SEI,

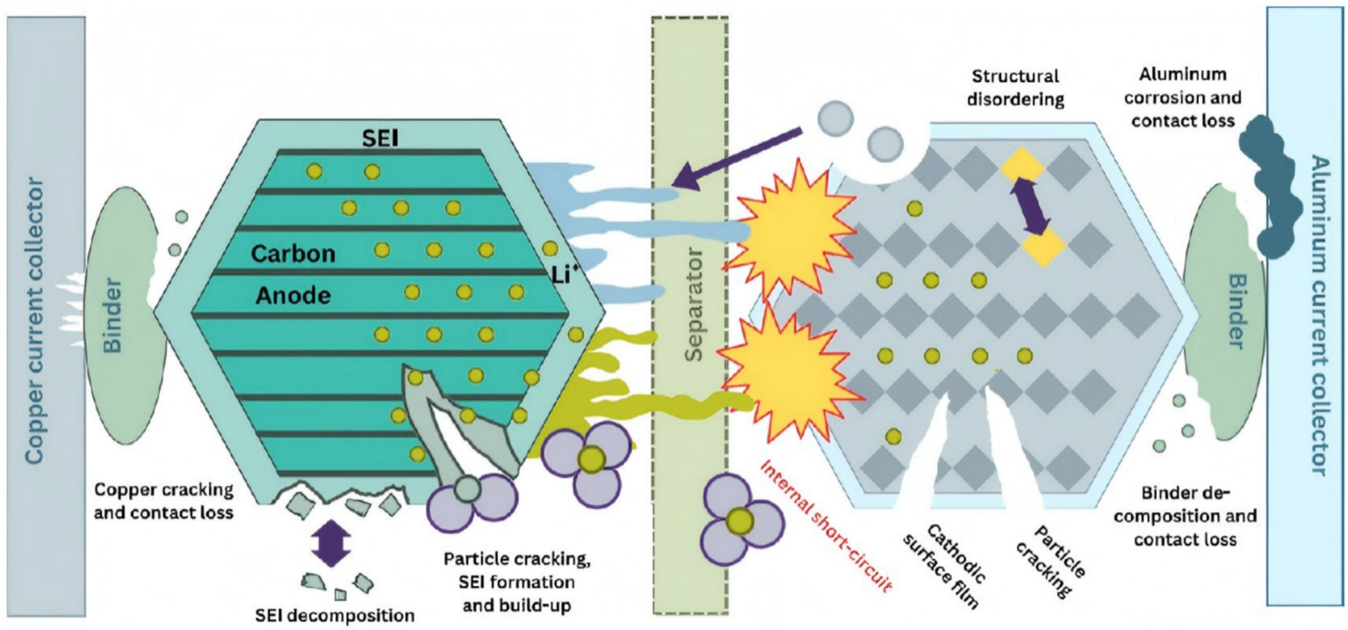

Fig. 5. Different types of degradation mechanics of a lithium-ion battery cell. Reproduced with permission from Ref. [212]. Copyright 2025, Elsevier.

loss of active lithium and electrode material, electrolyte decomposition, and particle cracking all contribute to increasing internal resistance and decreasing usable capacity. As these mechanisms accumulate, the cell exhibits gradual capacity fade, power loss, and a narrowing of its safe operating window. To study and model this degradation behaviour at scale, several groups have curated publicly available battery aging datasets that record long-term charge-discharge behaviour under controlled protocols. These shared resources substantially reduce experimental overhead, enabling methodological development without the need to assemble bespoke datasets, since generating large aging records typically requires prolonged cycling campaigns and considerable labour. Nevertheless, such datasets have inherent limitations: laboratory protocols are usually rigid and only approximate in-service duty cycles, and models trained solely on them may generalize poorly across chemistries, temperature ranges, or load profiles. This degradation complexity explains why the benchmark value of a dataset depends not only on its size, but also on whether it records degradation-relevant signals, diagnostic measurements, operating conditions, and EOL information.

NASA PCoE Li-ion aging dataset. The NASA Prognostics Center of Excellence (PCoE) Li-ion aging dataset was collected on commercially available 18,650 rechargeable cells using a custom battery prognostics test bed [39,40]. Each cell was repeatedly exercised under three operational profiles, namely charging, discharging, and electrochemical impedance spectroscopy (EIS), and the tests were conducted at multiple ambient temperatures in an environmental chamber. During discharge, specified current load levels were applied until the cell voltage reached preset cutoff thresholds, and some thresholds were set below the manufacturer-recommended 2.7 V limit in order to induce deep-discharge aging effects. The experiments were terminated at end of life, defined as a 30% fade in rated capacity, decreasing from 2 to 1.4 Ah. Measurements were acquired through a PXI-chassis-based data acquisition and control system at an approximate sampling rate of 10 Hz. The released files are organized by cycle and provide time-resolved trajectories of terminal voltage, current, and temperature for charge and discharge, together with discharge capacity. For impedance cycles, the dataset further reports processed impedance quantities, including battery impedance, rectified impedance, and fitted resistance terms R_e and R_{ct}.

In its original release, this dataset was intended to support battery prognostics research under varying load and thermal conditions, with particular emphasis on tracking degradation evolution to end of life and exploiting impedance-informed measurements for health assessment [39,40]. For subsequent machine-learning studies, its relatively small cohort but comparatively rich cycle-resolved and impedance-aware measurements make it especially suitable for feature-engineered and small-data settings, including models based on SVR, Gaussian process regression (GPR), and tree, as well as uncertainty-aware or physics-guided pipelines that can benefit from EIS-related variables and explicit degradation descriptors. By contrast, although the dataset remains widely used as a benchmark for deep-learning studies, its limited cohort size means that reported gains of highly parameterized models should generally be interpreted with appropriate caution.

MIT fast-charging dataset. The MIT fast-charging dataset contains measurements from 124 commercial 18,650 lithium iron phosphate cells with graphite anodes. The cells have a nominal capacity of 1.1 Ah and were cycled in a temperature-controlled chamber held at 30 °C [14]. To induce a wide range of lifetimes, each cell was charged from 0% to 80% state of charge using one of 72 single-step or two-step fast-charging policies, spanning average charging rates from 3.6 to 6 C. After charging, all cells followed the same discharge protocol, which applied a 4C current until the terminal voltage reached 2.0 V. End of life was defined as the cycle number at which the discharge capacity fell to 80% of its nominal value. During cycling, voltage, current, cell can temperature, and internal resistance were continuously recorded, and the release aggregates approximately 96,700 cycles across the full cohort.

In its original publication, this dataset was designed not only to generate a broad lifetime distribution under fast-charging conditions, but also to support early prediction of battery cycle life from features extracted from the initial cycles [14]. As such, it has become one of the most influential benchmark datasets for data-driven battery prognostics. For subsequent machine-learning studies, its medium-sized cohort, policy diversity, and long cycle-life span make it particularly suitable for early-life prediction, protocol-sensitive generalization analysis, and sequence-based learning from charge/discharge trajectories. It is therefore well matched to feature-based methods such as SVR, GPR, and tree ensembles, while also supporting deep sequence models such as CNN–, LSTM-, and Transformer-based architectures. At the same time, because the dataset is collected under a highly structured fast-charging design rather than broad real-world operating variability, reported gains on this benchmark should not be interpreted as direct evidence of universal robustness across chemistries or duty cycles.

University of Oxford Battery Degradation Dataset 1 (OBD-1). The Oxford Battery Degradation Dataset 1 reports long-horizon cycling experiments on eight small lithium cobalt oxide pouch cells tested in a temperature-controlled chamber maintained at 40 °C. The cells follow a constant-current charge stage followed by a constant-voltage hold, and they are then discharged under a variable-current load trace derived from the Artemis Urban driving cycle. To quantify capacity fade and diagnostic evolution over life, characterization measurements are conducted at regular intervals, with a characterization sequence recorded after every 100 drive-cycle runs. These characterization sequences include 1 C charge and discharge cycles and pseudo open-circuit voltage measurements. The released Matlab files organize records by cell and by characterization interval and provide time-resolved voltage, current, charge throughput, and temperature trajectories, together with an example file that illustrates the applied drive-cycle load profile [41,42].

This dataset was originally developed to study battery degradation under realistic drive-cycle-like loading while preserving regular characterization measurements that make long-term aging trajectories easier to interpret [41,42]. For subsequent machine-learning research, its small cohort but long-horizon cycling records and periodic characterization data make it particularly suitable for sequence-based degradation forecasting, feature extraction from drive-cycle responses, and studies that combine operational trajectories with diagnostic information. It is therefore especially useful for methods that can exploit structured temporal information under limited sample size, including feature-engineered regressors, recurrent models, and hybrid pipelines that incorporate characterization-aware health indicators. At the same time, because the dataset contains only a very small number of cells, results obtained on it should generally be interpreted as evidence of methodological feasibility rather than of broad statistical generalization.

CALCE battery aging dataset. The CALCE dataset hosts multiple lithium-ion datasets that cover cycling, storage, and diagnostic measurements, and it is widely used for SOH and RUL modeling. A representative subset is the accelerated cycle life testing dataset, which reports $LiCoO_2$-graphite pouch cells with a 3360 mA h rating, a 4.4 V charge cut-off, and a 3.0 V discharge cut-off. The experiments are conducted following a full-factorial design with three stress factors, namely ambient temperature set to 10, 25, 45, or 60 °C, discharge rate set to 0.7 C, 1 C, or 2 C, and charge cut-off current set to C/5 or C/40. Each cell is first operated using a standardized charging routine and a 0.7 C discharge to 3.0 V. After this initial procedure, cells are cycled under a stress condition, and a characterization capacity measurement is repeated after certain cycles. The released files are organized into initial characterization, continuous cycling, and periodical characterization tables, providing time-resolved voltage, current, temperature, and capacity records suitable for degradation and RUL research [43,44].

The original experimental design aimed to support controlled degradation studies under systematically varied stress conditions, thereby enabling researchers to examine how temperature, discharge rate, and charge termination jointly influence aging trajectories and health evolution [43,44]. For subsequent machine-learning studies, this structured factorial design makes the dataset particularly suitable for stress-aware prognostics, feature-engineered SOH/RUL modeling, and analyses of model robustness under changing operating conditions. It is therefore especially well matched to methods that can exploit compact descriptors or explicitly account for condition variation, including SVR-, GPR-, and tree-based models, as well as hybrid pipelines that incorporate diagnostic or physics-informed features. At the same time, because the broader CALCE resource consists of multiple subsets with differing protocols and organizational structures, careful subset selection and preprocessing are often required before meaningful cross-study comparison or unified model training can be carried out.

Stanford EV real-driving profiles dataset. This Data in Brief release reports aging data for ten INR21700-M50T lithium-ion cells with graphite and silicon in the anode and an NMC cathode, collected over 23 months in a temperature-controlled environment at 23 °C. Each aging cycle applies the Urban Dynamometer Driving Schedule as the discharge profile, and charging follows a constant-current then constant-voltage protocol with rates spanning from C/4 to 3 C. Degradation is periodically assessed through reference performance tests that include capacity tests, hybrid pulse power characterization, and EIS. The dataset provides time-aligned trajectories of voltage, current, and can temperature together with diagnostic measurements, enabling RUL studies that define end of life based on the reported capacity-fade curves under an application-specific threshold such as 80% of nominal capacity [45].

In its original release, this dataset was designed to capture battery aging under electric-vehicle-like duty cycles rather than under highly simplified laboratory protocols, while also preserving periodic diagnostic measurements that support mechanistic interpretation [45]. For subsequent machine-learning research, its main value lies less in cohort size than in its relatively realistic usage profile and multimodal diagnostic content. It is therefore particularly suitable for scenario-aware prognostics, multimodal feature engineering, and validation of methods intended to operate under application-relevant duty cycles, including hybrid models, diagnostic-informed regressors, and sequence models tested under more realistic discharge patterns. At the same time, because the dataset contains only ten cells, it is less appropriate as a standalone basis for claims of broad statistical generalization, and is more convincingly used for realistic-duty-cycle validation, cross-dataset transfer studies, or diagnostic-enhanced prognostics.

Potential value of newly released battery datasets. Beyond the five benchmark datasets discussed above, the public ecosystem of lithium-ion battery aging data has expanded markedly in recent years. Recent reviews, repositories, and newly released open datasets indicate that public resources are no longer confined to a few canonical benchmarks, but now span a broader and more heterogeneous landscape of aging data. These newer datasets cover wider variations in chemistry, operating conditions, protocol design, and diagnostic modalities, including large-scale cycling campaigns, combined calendar-cycle aging, and more application-oriented usage scenarios. Accordingly, their roles should be interpreted in a task-dependent manner: some of these resources have already become widely used tools for specific RUL-related tasks, whereas others provide complementary evidence for cross-dataset validation, transfer learning, and scenario-aware prognostics.

Several recent releases are particularly noteworthy because they expand the scale, diversity, and diagnostic richness of publicly available battery aging data, as summarized in Table 1. These datasets differ substantially in size, ranging from the 10-cell EV real-driving aging dataset reported by Pozzato et al. [45] to larger cohorts such as the 55-cell open-access run-to-failure dataset with unified preprocessing and benchmark baselines reported by Wang et al. [46], the 225-cell NMC/graphite lifetime dataset under a wide

Table 1
Representative recently released public battery aging datasets that extend the data foundation for RUL-related research.

Dataset/study	Year	Dataset size	Cathode/anode chemistry and setting	Main value	Refs.
Wang et al. open-access run-to-failure dataset	2024	55 cells	18,650 NCM523 lithium-ion cells; cathode: $LiNi_{0.5}Co_{0.2}Mn_{0.3}O_2$; anode: graphite; six charge/discharge strategies at 20 °C	Unified preprocessing and baselines for reproducible benchmarking	[46]
Pozzato et al. EV real-driving aging dataset	2022	10 cells	LG Chem INR21700-M50T cells; cathode: NMC/Li(Ni,Co,Mn)O_2; anode: graphite-silicon; UDDS-based discharge with CC–CV charging, HPPC, and EIS diagnostics	Real-driving-style duty cycles with multimodal diagnostics	[45]
Li et al. lifetime dataset under varying operating conditions	2024	225 cells	NMC/graphite lithium-ion cells; cycling under varied charge rates, discharge rates, and depth-of-discharge conditions	Larger public cohort for early-life and cross-condition prediction	[47]
Stroebl et al. multi-stage aging dataset	2024	279 cells	Samsung INR21700-50E cells; cathode: NCA / Li(Ni,Co,Al)O_2; anode: graphite-silicon; combined calendar and cycle aging across 71 conditions	Broad protocol diversity for transfer and scenario-aware studies	[48]
Stanford/Chueh-group aging matrix dataset	2025	359 cells	Commercial cylindrical 21,700 cells; cathode: Li(Ni,Co,Al)O_2; anode: graphite + SiO_x; 207 cycling protocols	Large protocol-diverse resource for cross-protocol generalization	[49]
Stanford/Chueh-group calendar aging dataset	2025	232 cells	Heterogeneous commercial Li-ion cells across eight cell types, four chemistries, and five manufacturers; cathode chemistries include LFP, NCA, NMC, and LCO; anode chemistry varies by cell type	Long-term calendar aging across chemistries, state of charges (SOCs), and temperatures	[50]
BatteryLife integrated benchmark	2025	16 datasets /990 batteries with life labels	Integrated multi-source battery-life benchmark; includes multiple battery types and chemical systems rather than a single cathode/anode pair	Large and diverse benchmark for cross-dataset evaluation	[51]

range of operating conditions reported by Li et al. [47], the 279-cell multi-stage aging dataset spanning 71 operating conditions reported by Stroebl et al. [48], and the recent Stanford/Chueh-group aging matrix dataset comprising 359 commercial 21,700 cells cycled across 207 distinct protocols [49]. The table also includes the 232-cell Stanford/Chueh-group calendar-aging dataset [50] and the BatteryLife integrated benchmark, which combines 16 datasets and 990 batteries with life labels [51]. Compared with earlier benchmark datasets, these resources provide broader coverage of operating conditions, larger cell cohorts, richer degradation trajectories, and in some cases more diverse diagnostic or protocol spaces. They therefore offer important new opportunities for cross-dataset validation, early-life prediction, transfer learning, and scenario-aware battery prognostics.

Nevertheless, the benchmark value of a battery aging dataset should not be judged by dataset size alone. For standardized RUL evaluation, several factors should be considered, including clearly specified cathode and anode chemistries, sufficient cohort size and cell-to-cell variability, well-documented charge/discharge and temperature protocols, complete or clearly defined lifetime trajectories, consistent end-of-life criteria, accessible metadata, and reproducible preprocessing and train/test splitting procedures. Recent public datasets have substantially expanded the empirical foundation of battery RUL research, and some of them have already become primary research tools in specific task settings, such as early-life prediction, fast-charging protocol evaluation, cross-condition prediction, and transfer learning.

At the same time, these datasets should not be treated as directly interchangeable benchmarks. Differences in cell chemistry, testing objectives, operating windows, diagnostic schedules, metadata organization, and end-of-life definitions can lead to different supervised learning targets and evaluation horizons. Therefore, rather than positioning one dataset as a universal standard for RUL evaluation, it is more appropriate to define the benchmark role of each dataset in a task-oriented manner. Some datasets are more suitable for early-life prediction, some for cross-condition validation, some for diagnostic-informed modeling, and others for testing transferability across chemistries or cycling protocols. This task-oriented perspective provides a more practical basis for dataset selection and for interpreting reported RUL prediction performance.

For researchers selecting datasets for ML-based RUL analysis, the intended task should be defined before model development. If the goal is to evaluate small-data regression, impedance-aware feature engineering, uncertainty-aware prediction, or physics-guided modeling, the NASA PCoE dataset is a suitable starting point because it provides charge, discharge, and EIS records under different temperatures. If the goal is early-life prediction or fast-charging protocol analysis, the MIT fast-charging dataset is more appropriate because it provides a larger standardized cohort of LFP/graphite cells cycled under diverse fast-charging policies. If the goal is feasibility-oriented sequence modeling under drive-cycle-like loading, the Oxford OBD-1 dataset can be useful, although its very small cohort limits broad statistical claims. If the goal is stress-aware modeling or condition-robustness analysis, CALCE is more suitable because it contains structured variations in temperature, discharge rate, and charge cut-off current. If the goal is realistic-duty-cycle validation or diagnostic-informed modeling, the Stanford EV real-driving profiles dataset is valuable because it combines EV-like discharge profiles with HPPC and EIS diagnostics, but it should not be used alone to claim broad generalization. For transfer learning, cross-condition evaluation, and cross-dataset benchmarking, recently released larger datasets and integrated resources such as the Li et al., Stroebl et al., Stanford/Chueh-group, and BatteryLife datasets are more appropriate, provided that chemistry, protocol, EOL definition, and train/test splitting are explicitly harmonized. This task-oriented guidance is summarized in Table 2.

4. Methods

This section reviews machine learning-based approaches for lithium-ion battery RUL prediction, focusing on data-driven and hybrid methods and examining them in terms of their core modeling ideas, typical input representations, applicable data regimes, and principal limitations. In a typical data-driven RUL workflow, degradation measurements collected from laboratory tests or field operation, including voltage, current, temperature, and capacity trajectories, are first preprocessed through synchronization, denoising, segmentation, and, where appropriate, feature construction. A learning model is then trained to approximate the mapping from input features or sequences to SOH or RUL targets, with val-

Table 2
Practical guidance for selecting public battery aging datasets for ML-based Li-ion battery RUL analysis.

Analysis objective	Recommended datasets	Selection rationale	Main caution
Small-data regression /uncertainty-aware modeling	NASA PCoE; CALCE	EIS, diagnostic records, stress-factor information, and compact health indicators support feature-engineered and physics-guided pipelines	Small cohorts or heterogeneous subsets; preprocessing choices can strongly affect comparability
Early-life prediction / fast-charging analysis	MIT fast-charging	124 LFP/graphite cells and 72 charging policies enable early-cycle feature learning and protocol-sensitive lifetime prediction	Structured fast-charging protocol; limited evidence for universal robustness across chemistries or real-world duty cycles
Drive-cycle-like sequence modeling	Oxford OBD-1; Stanford EV	Drive-cycle-like loading, long-horizon trajectories, and periodic characterization support temporal and hybrid sequence models	Very small cohorts; results are better interpreted as feasibility evidence rather than broad statistical generalization
Stress-aware/cross-condition modeling	CALCE; Li et al.; Stroebl et al.	Varied temperature, charge/discharge rate, DoD, charge termination, and multi-stage aging conditions support robustness analysis	Protocol, EOL, metadata, and train/test split harmonization are required for fair comparison
Realistic-duty-cycle validation	Stanford EV	EV-like discharge profiles with HPPC and EIS diagnostics support scenario-aware and diagnostic-informed validation	Only ten cells; not sufficient alone for broad generalization claims
Transfer learning/ cross-dataset bench-marking	Recent larger datasets; BatteryLife	Broader chemistry, protocol, operating-condition, and data-source diversity support transfer and cross-dataset evaluation	Chemistry, protocol, EOL definition, metadata, and split differences must be explicitly handled

idation and hyperparameter tuning used to balance predictive accuracy and generalization. The overall process is summarized in Fig. 6, and the following subsections further examine how different input representations and model families shape this mapping.

In what follows, this review first summarizes the main input features and data representations used in battery RUL prediction, thereby providing a clearer basis for the subsequent methodological discussion. It then reviews traditional and shallow machine-learning methods and deep neural architectures in turn, before moving to ensemble data-driven methods and, finally, hybrid approaches that combine learning-based predictors with physical or mechanistic constraints.

4.1. Input features and data representations for battery RUL prediction

The performance of battery RUL models depends not only on the learning architecture, but also on how degradation information is represented before it is presented to the model. Recent battery-ML workflows therefore treat data preprocessing, feature extraction, and target construction as an explicit stage rather than a purely auxiliary step [52,53]. In practice, the input space of lithium-ion battery prognostics is not organized by algorithm alone; instead, several recurring data representations are shared across different model families, with each family exhibiting different preferences in terms of dimensionality, temporal structure, interpretability, and data demand.

Raw telemetry and partial-cycle sequences. The most direct representation consists of raw or lightly processed voltage, current, temperature, and capacity trajectories, either over full cycles or over diagnostically informative partial charge/discharge segments. These inputs preserve rich temporal information and are therefore naturally suited to sequence-oriented deep models, including CNNs, TCNs, LSTM networks, gated recurrent units (GRUs), and Transformer-based architectures, which can learn local motifs, temporal dependencies, or long-range cross-channel relations directly from the signal stream [52]. Their main advantage is reduced reliance on manual feature design; however, they typically require stronger preprocessing, more training data, and greater care in handling sequence length, alignment, and noise.

Engineered health indicators and compact descriptors. A second and still widely used representation is based on engineered health indicators and low-dimensional descriptors, such as capacity fade, internal resistance, charge/discharge duration, relaxation characteristics, slope statistics, and cycle-wise summary variables. These features compress degradation information into compact tabular inputs and remain especially attractive for SVR, GPR, tree ensembles, shallow multilayer perceptrons (MLPs), and ELM-type models, which often perform well in small- to medium-sized datasets when the input variables are physically meaningful and well curated [54,55]. Their main strengths are interpretability, data efficiency, and ease of deployment, whereas their main limitation is the dependence on feature engineering choices, which may constrain transferability across datasets and operating conditions.

Mechanism-relevant diagnostic features. Beyond generic statistical descriptors, many studies rely on mechanism-relevant diagnostic features, particularly incremental-capacity/differential-voltage (IC/DV) curves, EIS, and related relaxation or pulse-derived indicators. These representations are attractive because they are more tightly coupled to electrochemical degradation processes and can therefore provide more informative health signals than raw telemetry alone [56,57]. In the RUL literature, such features are commonly paired with GPR, tree models, SVR, and hybrid or physics-informed pipelines, although they can also be fused with deep encoders in multimodal settings. Their main limitation is that diagnostic acquisition is often sparse, protocol-dependent, and not yet fully standardized.

Structured and multimodal representations. For module- and pack-level prognostics, the input is often not a single trajectory but a structured collection of interacting measurements. In such cases, cell-level telemetry, voltage segments, or sensor streams can be represented as graphs, with edges encoding similarity, electrical coupling, or thermal interaction. This representation is particularly suitable for graph neural networks (GNNs), while hybrid frameworks may further combine telemetry, diagnostic signals, and physics-derived latent states in a unified model [52,58]. Compared with single-stream inputs, structured and multimodal representations offer greater expressive power for capturing interaction effects, but they also introduce additional design choices regarding graph construction, feature fusion, and data consistency.

Overall, input representation should be viewed as a first-order design decision in battery RUL prediction, rather than a secondary implementation detail. The main input-representation categories discussed above, together with their typical alignment with model families, advantages, and limitations, are summarized in Table 3. In the following subsections, different algorithm families are therefore discussed together with the input regimes to which they are

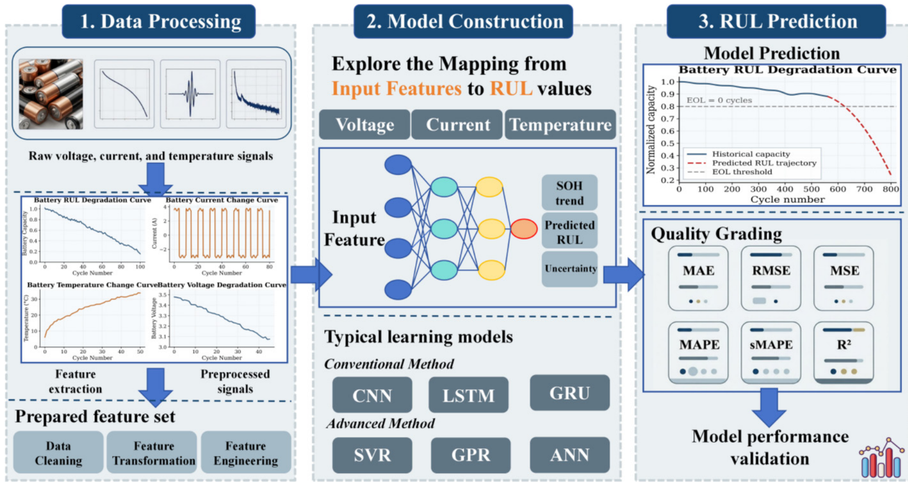

Fig. 6. Data-driven RUL prediction framework for lithium batteries.

most naturally suited, instead of being interpreted as architecture-only choices.

4.2. Traditional and shallow machine learning methods

Traditional and shallow machine learning methods formed the first major wave of battery RUL prediction and remain widely used as baselines. As summarized in Table 4, this section focuses on representative families that remain methodologically important in the battery RUL literature, namely support-vector methods, tree ensemble models, shallow neural networks and ELM, pipelines based on autoencoders (AEs), and GPR. They are attractive in data-scarce or resource-constrained settings because they train quickly, can work with relatively small datasets, and are often easier to tune and interpret than deep architectures. However, their performance depends heavily on feature engineering and they may struggle to capture highly nonlinear, condition-dependent degradation once telemetry becomes high dimensional. Rather than treating these approaches as uniformly competitive, the following subsections examine where they remain effective, what kinds of inputs and dataset regimes they are best suited to, and why their reported performance should be interpreted in light of feature design, preprocessing choices, and evaluation settings.

4.2.1. Support-vector methods

Support-vector methods were originally developed for large-margin classification with support-vector machines and later extended to regression with support-vector regression, forming a

Table 3
Common input representations for Li-ion battery RUL prediction and their typical alignment with model families.

Representation	Typical examples	Commonly paired methods	Main advantages	Main limitations
Raw telemetry and partial-cycle sequences	Voltage, current, temperature, and capacity trajectories; full-cycle or partial-cycle charge/discharge segments	CNN/TCN, LSTM/GRU, and Transformer-based models	Preserve rich temporal information; enable end-to-end feature learning; reduce reliance on manual feature engineering	Require stronger preprocessing and larger datasets; sensitive to sequence alignment, variable length, and measurement noise
Engineered health indicators and compact descriptors	Capacity fade, internal resistance, charge/discharge duration, relaxation descriptors, slope statistics, and cycle-wise summary variables	SVR, GPR, tree ensembles, shallow MLPs, and ELM-type models	Interpretable, data-efficient, and easy to deploy; suitable for small-to medium-sized datasets	Depend strongly on Feature engineering quality; may limit transferability across datasets and operating conditions
Mechanism-relevant diagnostic features	Incremental-capacity/differential-voltage (IC/DV) curves, electrochemical impedance spectroscopy (EIS), and relaxation- or pulse-derived indicators	GPR, SVR, tree models, hybrid and physics-informed pipelines; multimodal deep encoders	More tightly coupled to electrochemical degradation mechanisms; often provide more informative health signals than raw telemetry alone	Diagnostic acquisition is often sparse, protocol-dependent, and not yet fully standardized
Structured and multimodal representations	Cell-level telemetry graphs, voltage-segment graphs, sensor networks, and telemetry fused with diagnostics or physics-derived latent states	GNNs, multimodal Transformer models, and hybrid frameworks	Capture interaction effects, topology, and cross-source information; suitable for module- and pack-level prognostics	Require additional design choices for graph construction, feature fusion, and data consistency; computationally more demanding

Table 4
Traditional and shallow machine learning methods for Li-ion battery RUL prediction.

Method family	Structure	Strengths	Limitations	Data/resource regime	Typical use cases	Refs.
SVR	Fig. 7	Strong accuracy on small-medium tabular features; robust margin-based regression with mature theory	Requires feature engineering; kernel and hyperparameter tuning are sensitive; scaling to very large datasets is costly	Typically small medium datasets (roughly < 50 to 50–150 cells); central processing unit (CPU) training and inference usually sufficient	Early RUL baselines using hand-crafted health indicators or summary statistics from cycling data	[213,214]
GPR	Fig. 7	Bayesian model with predictive mean and variance; naturally provides calibrated uncertainty for risk-aware decisions	Cubic complexity in sample size; kernel choice is critical; not suitable for very large fleets without sparse approximations	Typically very small–small datasets (usually < 50 cells) where uncertainty quantification is essential	Data-scarce, safety–critical RUL estimation with moderate horizons	[94,215]
Tree ensembles	Fig. 7	Strong tabular baselines; capture nonlinearities and feature interactions; robust and fast on standard CPUs	Depend on engineered features; limited ability to exploit detailed temporal structure; weak extrapolation outside training range	Small to large tabular datasets (from <50 to >150 cells); efficient CPU training and deployment	RUL prediction from summary statistics or Health indicators extracted from cycling curves	[90,216]
MLP	Fig. 7	Simple and fast; approximate nonlinear mappings from compact feature sets to RUL with minimal implementation effort	Limited depth and expressiveness compared with deep models; performance depends on feature quality and regularization	Typically small–medium datasets (roughly <50 to 50–150 cells) with low-dimensional indicators; very low compute and memory requirements	Baseline RUL regressors on engineered health indicators or reduced-order descriptors	[217,218]
ELM	Fig. 7	Extremely fast and lightweight training with closed-form output layer; low data and resource demand	Random hidden layer can yield variable performance; shallow capacity; interpretability is modest	Typically small–medium datasets (roughly <50 to 50–150 cells); suitable for rapid retraining, online updating, and edge deployment	Fast RUL estimation, including OS-ELM-style variants for streaming degradation	[78,219]
AE-based	Fig. 7	Unsupervised feature learning and denoising; compress raw trajectories into compact health indicators	Need sufficient data for meaningful latent spaces; reconstruction loss may not perfectly align with RUL accuracy	Typically small-medium time-series datasets (roughly <50 to 50–150 cells); moderate graphics processing unit (GPU) or CPU resources suffice	Pretraining health indicators from voltage-current time series before shallow or deep RUL regressors	[220,221]

unified family of kernel-based learners that is attractive for battery prognostics because it combines data efficiency with flexible nonlinear representations [59–61]. In battery RUL studies, classification-oriented support vector machines (SVMs) are mainly used to delineate health or degradation regimes, whereas SVR is more commonly adopted as a regression model for compact health indicators or other low-dimensional descriptors derived from cycling data. As shown in Fig. 7, the core idea of SVR is to learn a nonlinear mapping while controlling model complexity through margin-based regularization. In its kernel form, the prediction function can be written as

$$f(x) = \sum_{i=1}^{n} (a_i^* - a_i) K(x_i, x) + b$$

where $K(x_i, x)$ is a kernel function that enables nonlinear regression in an implicit high-dimensional feature space. Together with the ε-insensitive loss, this formulation makes SVR relatively robust to small perturbations and well suited to small-sample regression problems.

In the battery RUL literature, SVR is most commonly applied to hand-crafted health indicators, summary statistics, or decomposed capacity signals, rather than to high-dimensional raw telemetry. This usage pattern reflects its principal strength in this domain: when degradation information has been compressed into physically meaningful tabular descriptors, SVR remains a strong and computationally efficient baseline [62,63]. In practical workflows, it is also frequently embedded in multistage pipelines in which regime identification, denoising, or signal decomposition precedes final RUL regression.

Subsequent studies have sought to extend SVR beyond this baseline role through three main directions. The first is hyperparameter optimization, for example PSO-SVR and related metaheuristic variants, which improve fit stability under limited data [64–66]. The second is decomposition-enhanced SVR, where methods such as CEEMDAN are introduced to suppress noise or capacity-regeneration fluctuations before regression [67,68]. The third is probabilistic or hybrid integration, in which SVR is coupled with filters, state-space models, or quantile formulations to provide more robust long-horizon prediction or interval-aware outputs [69–72].

From a critical perspective, however, the reported performance of SVR should be interpreted with care. In many cases, performance gains arise not from the regressor alone, but from the quality of the input features, the preprocessing or decomposition scheme, and the hyperparameter-search strategy. This means that SVR remains highly competitive in small-to medium-sized, feature-engineered settings, but is generally less suitable when the task requires direct representation learning from rich, high-dimensional telemetry. Its main limitations therefore lie in its reliance on feature engineering, sensitivity to kernel and tuning choices, and limited scalability relative to modern deep architectures.

As shown in Fig. 8(a), SVR-based methods accounted for more than 20%–30% of RUL studies in the early years of this research trend (2017–2018). At that stage, their algorithmic simplicity and relatively low data requirements made them particularly attractive to researchers. However, the share of SVR has gradually declined over time—both among hybrid approaches and across all meth-

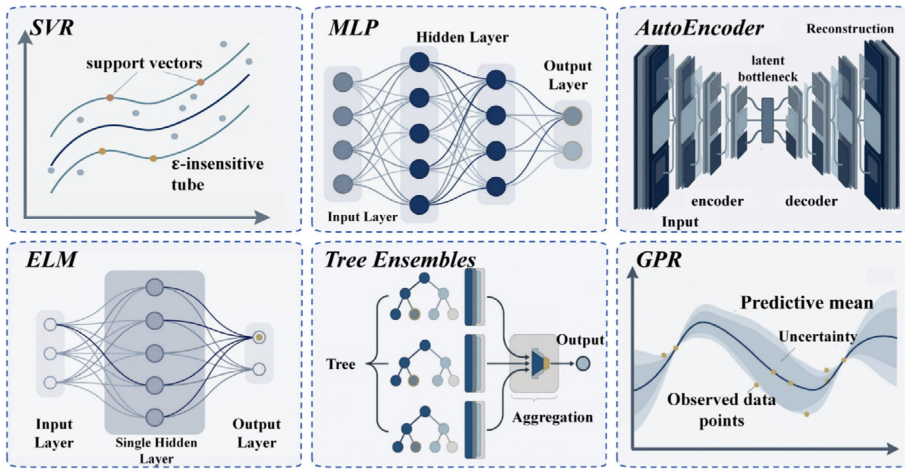

Fig. 7. Structure of traditional RUL prediction methods.

ods—and in recent years its usage has dropped to below 10% of the surveyed literature. This decline does not imply that SVR has lost methodological relevance; rather, it indicates that its role has shifted from a dominant standalone predictor to a strong baseline, a small-data regressor, and a lightweight component in more elaborate prognostic pipelines.

4.2.2. Shallow neural networks: MLP, ELM, and autoencoder-based models

Shallow neural architectures remain relevant in battery RUL prediction because they occupy an intermediate position between purely kernel-based regressors and deeper sequence models. In this review, we group under this family feed-forward MLP, shallow autoencoder-based pipelines that learn compact representations of degradation signals, and ELM and their variants. Although these methods differ in training strategy, they share a common practical role in the literature: they are most often used when the input has already been compressed into engineered health indicators, short sliding-window descriptors, or other relatively low-dimensional representations, and when computational efficiency or limited data availability makes deeper architectures less attractive.

Among these models, shallow MLPs provide the most direct supervised mapping from input descriptors to RUL. A typical one-hidden-layer formulation can be written as

$$\hat{y} = W^{(2)}\sigma\left(W^{(1)}x + b^{(1)}\right) + b^{(2)}$$

where x is the input feature vector and $\sigma(\cdot)$ is a nonlinear activation function. In battery RUL studies, such models are usually applied to engineered health indicators or short cycle-level summaries of voltage, current, and temperature, rather than to long raw telemetry streams. Their main attraction is simplicity: they are easy to train, computationally light, and often sufficient when the relationship between selected features and remaining life is nonlinear but not highly structured. Representative studies therefore use shallow MLPs as lightweight regressors, sometimes with Bayesian variants or limited temporal concatenation to improve robustness without introducing the full complexity of recurrent models [73].

Autoencoder-based pipelines play a somewhat different role. Rather than serving primarily as end predictors, shallow AEs are most often used as front-end representation learners that compress noisy cycling signals into lower-dimensional latent codes. This can be expressed through the standard reconstruction objective

$$\min_{\phi,\psi} \|x - \psi(\phi(x))\|_2^2$$

where the encoder $\phi(\cdot)$ maps the input to a latent representation and the decoder $\psi(\cdot)$ reconstructs the original signal [74]. In the battery RUL literature, stacked, denoising, and related shallow autoencoder variants are typically used to extract compact health indicators before regression, or to provide noise-robust latent features for subsequent shallow or sequential predictors [75–77]. Their practical value therefore lies less in standalone forecasting accuracy than in improving representation quality when raw degradation signals are noisy, redundant, or only weakly aligned with the final RUL target.

ELM constitute the most lightweight branch within this family. ELM formulates regression with a single hidden layer whose input

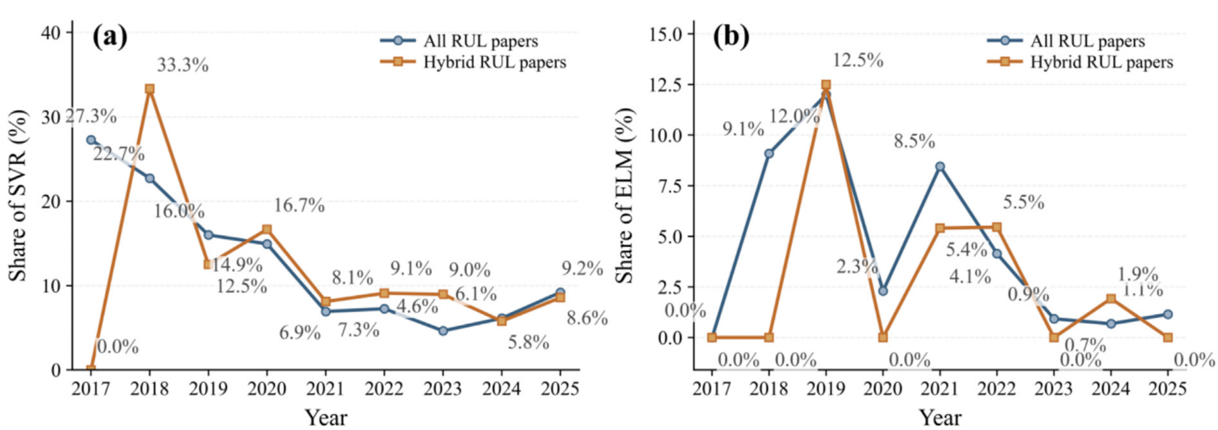

Fig. 8. (a) Trends of SVR adoption; (b) trends of ELM adoption.

weights and biases are randomly assigned, while the output weights are solved analytically in closed form. A standard ELM predictor can be written as

$$\hat{y} = \sum_{i=1}^{L} \beta_i g(w_i x + b_i)$$

where the hidden parameters are fixed random quantities and the output weights are obtained from a least-squares solution. In battery prognostics, this design is attractive because it greatly reduces training cost and enables rapid model updating, which explains why online variants such as OS-ELM have been explored for streaming SOH/RUL estimation [78,79]. Subsequent studies have attempted to mitigate the instability of random hidden-node assignment through multiple-kernel designs, deep or stacked ELM variants, and *meta*-heuristic optimization, while some pipelines further combine ELM with sparse Bayesian regression to provide interval-aware outputs [80–85].

From a comparative perspective, these three branches should not be viewed as equally strong substitutes for modern deep architectures. Shallow MLPs are best understood as efficient nonlinear regressors for compact descriptors; AEs are primarily useful as representation-learning front ends; and ELM is attractive mainly when very fast training or lightweight deployment is required. Reported performance improvements within this family are therefore often tied less to architectural novelty than to feature quality, denoising strategy, latent representation design, or hyperparameter optimization. In particular, once the task shifts from compact tabular descriptors to rich, high-dimensional, and strongly condition-dependent telemetry, these shallow models generally become less competitive than architectures that can learn hierarchical or long-range temporal representations directly.

As shown in Fig. 8(b), the adoption of ELM has remained limited and has gradually declined in recent years. Although its share once peaked at around 12% in 2019, it has since dropped to around 1% in the most recent literature. This trend reflects a broader shift in the field: as datasets have become richer and model expectations have moved toward stronger representation learning and cross-condition robustness, the role of shallow neural methods has evolved from that of dominant predictors to that of lightweight baselines, feature compressors, or computationally efficient components within larger prognostic pipelines.

4.2.3. Tree ensemble-based RUL prediction

Tree ensemble methods occupy a practical and still important position in battery RUL prediction because they are particularly well suited to tabular, engineered inputs with limited sample size and nonlinear feature interactions. In contrast to shallow neural models, which still rely on learned latent mappings, tree ensembles operate directly on explicit descriptors and therefore remain attractive when the feature set has already been condensed into health indicators, summary statistics, or mechanism-relevant diagnostics. Within this family, RFs emphasize variance reduction through aggregation, whereas boosting-based methods such as GBDT, XGBoost, and LightGBM emphasize sequential error correction and stronger fitting capacity.

At the level of a single learner, a regression tree maps an input x to a piecewise-constant prediction by partitioning the feature space into disjoint regions

$$h(x) = \sum_{j=1}^{J} b_j \, 1_{R_j}(x)$$

where b_j is the leaf prediction associated with region R_j. Building on this idea, RFs average predictions across many decorrelated trees, whereas boosting-based methods update the ensemble stage by stage, for example through

$$F_m(x) = F_{m-1}(x) + \gamma_m h_m(x)$$

so that each new tree focuses on the residual error left by the current model. For battery RUL tasks, these formulations are less important as mathematical objects in themselves than as indicators of the practical strengths of tree ensembles: they can capture nonlinear interactions, tolerate heterogeneous feature scales, and often train efficiently on standard CPU hardware. In the battery prognostics literature, tree ensembles are most commonly applied to engineered feature sets rather than to raw telemetry. Representative inputs include cycle-level summary statistics, health indicators derived from voltage-current–temperature trajectories, and physically informed descriptors such as incremental-capacity, differential-voltage, and relaxation features. In this regime, RFs provide robust baselines with limited tuning burden, while boosting-based models are often favored when stronger fitting power is needed. XGBoost, for instance, has been used as the regression head within hybrid Bayesian filtering frameworks to improve both predictive accuracy and uncertainty handling [86]. Pure boosting pipelines with sliding-window features have likewise shown competitive RUL performance under varying conditions [87], and LightGBM- or RF-based approaches have further benefited from *meta*-heuristic hyperparameter optimization for improved robustness across cells and operating regimes [88].

From a performance perspective, the literature generally suggests that tree ensembles remain highly competitive on small- to medium-sized datasets, especially when the feature space already encodes degradation-relevant structure. Under such conditions, boosted-tree models can approach, and in some studies rival, deeper neural networks while retaining lower training cost and stronger interpretability at the feature level [89]. Their reported gains are particularly credible when the inputs incorporate physically meaningful descriptors, because in that case the model is exploiting both nonlinear regression capacity and informative feature design. This is also why random-forest variants trained on battery-aging descriptors have been successfully extended from SOH estimation to RUL prediction through trend extrapolation [90].

A critical point, however, is that the performance of tree ensembles in battery RUL prediction is often inseparable from the quality of the engineered feature set. Their apparent strength does not usually arise from end-to-end representation learning, but from their ability to make efficient use of compact and informative descriptors. As a result, these methods are best understood as strong classical baselines for structured tabular regimes, rather than as universally superior predictors. Once the task shifts toward long raw sequences, richer multimodal telemetry, or strongly condition-dependent temporal dynamics, tree ensembles generally become less competitive than architectures that can learn hierarchical and sequential representations directly. Their main limitations therefore lie in their dependence on feature engineering, weaker handling of fine-grained temporal structure, and limited extrapolation outside the range represented in the training data.

4.2.4. Gaussian process regression

GPR provides a nonparametric Bayesian framework for regression that not only predicts a mean response but also quantifies predictive uncertainty. A Gaussian process is defined as a collection of random variables, any finite subset of which has a joint Gaussian distribution

$$f(x) \sim \mathcal{GP}(m(x),\, k(x, x'))$$

where $m(x) = \mathbb{E}[f(x)]$ is the mean function and $k(x, x') = \mathrm{Cov}[f(x), f(x')]$ is the covariance function. Given training inputs $X = \{x_i\}_{i=1}^{n}$ and corresponding outputs $y = \{y_i\}_{i=1}^{n}$, the joint prior between observed outputs and predictive outputs $f*$ at test inputs $X*$ remains Gaussian; conditioning on the observed data yields an analytic posterior with a predictive mean and variance that reflect both fit and uncertainty.

The kernel function plays a central role in GPR, defining the similarity between inputs and thereby determining the structure of the covariance matrix used in prediction. Common kernel choices such as the radial basis function or Matérn kernels encode smoothness and can model nonlinear patterns. Hyperparameters of the kernel and noise variance are typically learned by maximizing the marginal likelihood of observed data, which balances model complexity and data fit. For battery RUL prediction, this probabilistic formulation is particularly valuable because the predictive variance can be interpreted as an explicit measure of confidence, which is useful when model outputs are intended to support risk-aware maintenance and operational decision-making.

In Li-ion battery RUL prediction, accurate estimation of future capacity degradation and remaining life critically depends on capturing complex nonlinear degradation dynamics from limited cycling data. GPR's kernel-based representation allows it to approximate such nonlinear capacity-cycle relationships effectively, and its predictive variance conveys calibrated uncertainty that is useful for risk-aware maintenance planning and decision-making. In practice, GPR is most often applied to engineered degradation features, health indicators, or compact representations of cycling behavior, rather than to long raw telemetry streams. This input regime explains why GPR remains especially attractive in small-data settings: when the feature space is already informative and the sample size is limited, GPR can provide a strong compromise between predictive accuracy and uncertainty awareness. This has motivated numerous recent applications of GPR to battery RUL tasks, where GPR is employed either directly on degradation features or as part of hybrid models that integrate feature decomposition or auxiliary predictors to enhance performance [91–93]. Reported performance gains of GPR should nevertheless be interpreted with some caution. In many studies, its effectiveness depends strongly on the quality of feature design, kernel specification, and preprocessing choices, rather than arising from the regression model alone. Accordingly, GPR is best understood not as a universally superior predictor, but as a particularly suitable choice when calibrated uncertainty, small-sample robustness, and physically meaningful tabular inputs are all important. However, the cubic scaling of standard GPR with the number of training points and the sensitivity of performance to kernel and hyperparameter design have led to work on sparse approximations and composite kernels in large battery datasets to retain uncertainty quantification while mitigating computational cost [94]. These limitations also mean that, once the task shifts toward richer telemetry, larger cohorts, or more strongly condition-dependent sequential degradation patterns, GPR often becomes less competitive than models that can learn hierarchical temporal representations more directly.

4.3. Deep and advanced neural architectures for RUL prediction

Over roughly the past decade, deep neural architectures have evolved from isolated case studies to central tools for battery RUL prediction. As summarized in Table 5, this section focuses on four representative families that have become particularly influential in the recent literature, namely convolutional networks (CNN/TCN), gated recurrent networks (LSTM/GRU), Transformer-based sequence models, and GNNs [95]. These architectures are especially attractive when degradation is strongly nonlinear and richer telemetry or structured measurements are available, often yielding improved representation learning and stronger cross-condition modeling capacity, albeit at the cost of greater data and computational demands and reduced interpretability [96]. The following subsections therefore emphasize not only their architectural characteristics, but also the conditions under which their reported performance gains are most credible and practically meaningful.

4.3.1. Convolutional neural network-based RUL prediction

CNN-based models, as illustrated in Fig. 9, are feed-forward architectures built from repeated blocks of convolution, nonlinear activation, optional normalization, and pooling or striding [97]. In battery RUL prediction, they are most commonly used when the input is available as raw or lightly processed one-dimensional sequences, such as capacity trajectories, voltage-current curves, or short sliding-window telemetry segments, and when the main objective is to learn degradation-sensitive local patterns directly from the signal stream. A key component of CNNs is the convolutional layer, which applies a set of learnable kernels to input data. For one-dimensional signals such as capacity or voltage time series, the convolution operation at layer l with a kernel $w^{(l)} \in \mathbb{R}^k$ can be written as

$$z_i^{(l)} = \sum_{j=0}^{k-1} w_j^{(l)} x_{i+j}^{(l-1)} + b^{(l)}$$

where $x^{(l-1)}$ is the input sequence from the previous layer, $z^{(l)}$ is the linear output (feature map), and $b^{(l)}$ is a bias. A nonlinear activation such as $\mathrm{ReLU}(z) = \max(0, z)$ is then applied element-wise, producing activated feature maps that highlight learned motifs.

Weight sharing across positions and local connectivity induce translation equivariance and substantially reduce the number of parameters compared with fully connected networks, enabling scalable learning of local patterns such as charge plateaus and knee points [98,99]. Pooling or striding further reduces the temporal resolution while increasing the receptive field, allowing stacked layers to build hierarchical representations from primitive motifs to higher-level degradation signatures. These properties explain why CNN-based models have become attractive in battery prognostics: they can extract informative local degradation motifs with relatively good parameter efficiency, without relying as heavily on manual feature engineering as traditional shallow methods.

These properties make CNNs a convenient front end for RUL prediction, either as standalone encoders or in combination with recurrent and attention-based modules for downstream regression. For example, time-domain CNN models treat capacity and voltage-current sequences as one-dimensional signals for local feature extraction, and the resulting representations can be passed to sequence models such as BiLSTM to propagate long-range dependencies for stable RUL forecasts on public benchmarks [100]. In parallel, TCNs replace recurrence with causal, dilated convolutions and residual blocks to enlarge the effective receptive field while preserving chronology, which has been shown to capture capacity-regeneration patterns and improve robustness under varying conditions [101]. Recent hybrid designs further combine signal decomposition with convolutional encoders and recurrent decoders, for example coupling CEEMDAN-based preprocessing with a CNN-RNN architecture to mitigate noise and nonstationarity and achieve consistent gains on NASA- and CALCE-type degradation datasets [102]. Taken together, the reported literature suggests that CNN-based methods are most competitive when degradation information is expressed through locally informative temporal structure and when the dataset is sufficiently rich to support end-to-end representation learning. Their reported gains are often especially convincing when the alternative would otherwise require extensive hand-crafted feature design.

From a critical perspective, however, the strengths of CNNs in battery RUL prediction should not be overstated. Their superiority is not universal, and reported improvements often depend on input

Table 5
Deep and advanced neural architectures for Li-ion battery RUL prediction.

Method family	Structure	Strengths	Limitations	Data/resource regime	Typical use cases	Refs.
CNN	Fig. 9	Learn local degradation patterns directly from time-domain signals; parameter sharing gives good data efficiency and scalability; TCNs use causal, dilated convolutions to preserve chronology	Mainly capture local-mid-range dependencies; very long-range memory requires deeper stacks and careful tuning; architecture and kernel size choices are dataset dependent	Typically medium-large regimes (roughly 50–150 to >150 cells), often with dense telemetry	End-to-end RUL prediction from raw or lightly processed voltage-current sequences where charge/ discharge plateaus, knees, and regeneration patterns are informative	[100,101]
LSTM	Fig. 9	Explicit temporal memory; handle long sequences and nonstationary trends; widely used baseline with well-understood training recipes for sequence modeling	Training can be slow on long histories; still vulnerable to gradient issues and irregular sampling if not carefully regularized; may overfit small datasets	Often medium-sized cohorts (roughly 50–150 cells) or smaller cohorts with long sequences	Cycle-level RUL forecasting from full charge discharge histories under varying operating profiles and load conditions	[222,223]
GRU	Fig. 9	Similar modeling capacity to LSTM with fewer parameters and simpler gating; often achieves comparable accuracy at lower computational and memory cost	Inherits recurrent nueral networks (RNNs) limitations on very long or highly irregular sequences; performance sensitive to depth and hidden size choices	Typically small-medium cohorts (roughly <50 to 50–150 cells)	RUL prediction where a lightweight recurrent model is needed to balance accuracy, latency, and power consumption	[224,225]
Trans.	Fig. 9	Self-attention captures long-range and cross-channel dependencies; naturally fuses heterogeneous signals and metadata; training is highly parallelizable on modern hardware	Data- and compute-hungry; higher risk of overfitting on small datasets; attention patterns can be hard to interpret and require regularization	Typically medium–large regimes (roughly 50–150 to >150 cells) with rich multivariate telemetry	Long-horizon, cross-condition RUL prediction on large and heterogeneous fleets with rich voltage-current–temperature histories	[118,226]
GNN	Fig. 9	Exploit pack topology, segment similarity graphs, or sensor networks; jointly model spatial and temporal dependencies for cell–cell interactions	Sensitive to graph construction; computational cost increases with graph size; topology may vary with operating conditions	Typically medium-large structured datasets (roughly 50–150 to >150 cells)	Pack- and module-level RUL estimation when cell interactions, spatial layout, and thermal coupling significantly affect degradation	[227,228]

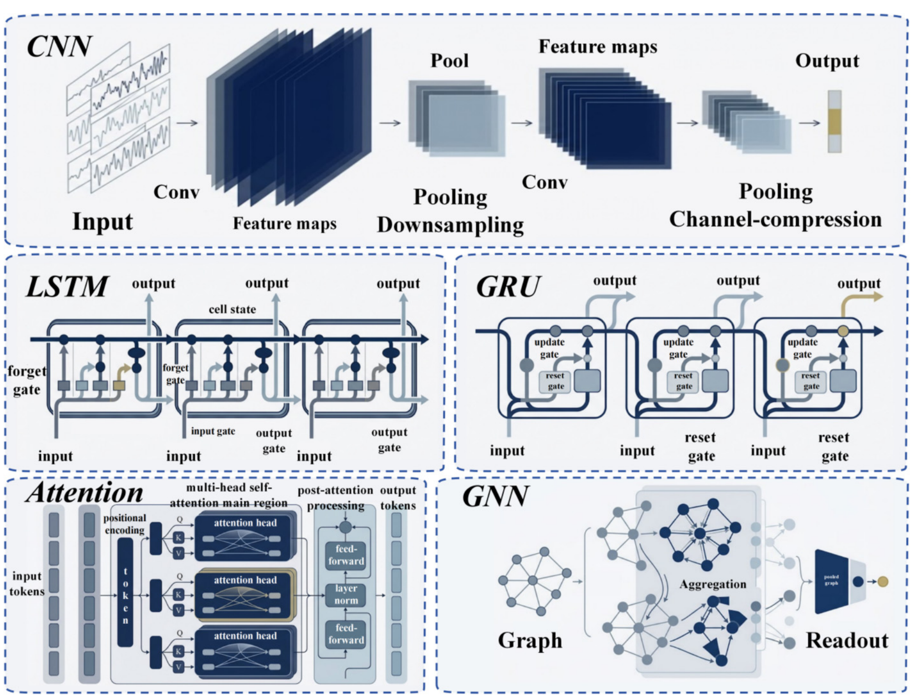

Fig. 9. Structure of advanced RUL prediction methods.

formatting, window design, denoising strategy, and whether convolution is used alone or as a front end for recurrent or attention-based back-end modules. Moreover, standard CNNs primarily emphasize local and mid-range temporal structure; when very long-range dependencies, strong condition shifts, or heterogeneous multimodal inputs dominate the task, their performance may depend heavily on additional architectural components such as dilation, recurrence, or attention. CNN-based models are therefore best understood as powerful representation learners for structured temporal signals, rather than as universally dominant predictors across all battery RUL settings.

As shown in Fig. 10(a), compared with earlier SVR- and ELM-based approaches, CNN-based methods are emerging as a rising star: their adoption has increased year by year, from essentially zero usage in the early stage to around 25% of the surveyed literature at present, with a clear upward trend. This growing popularity is largely driven by the ability of CNNs to automatically extract local degradation patterns from raw voltage-current–temperature time series, their parameter sharing and receptive-field structure which make them relatively data-efficient and scalable, and their compatibility with hybrid architectures such as CNN-RNN or CNN-attention models [103,104].

As larger and more complex datasets become available, these advantages make CNN-based models increasingly attractive compared with traditional shallow learners such as SVR and ELM, particularly in scenarios that require robust feature learning and end-to-end modeling of nonlinear degradation dynamics.

4.3.2. Gated recurrent networks for RUL prediction

Recurrent architectures model battery degradation as a time-ordered sequence of feature vectors, for example capacity or voltage-current summaries over successive cycles or sliding windows, allowing the predictor to integrate both short-range and long-range dependencies. In battery RUL prediction, they are most naturally suited to settings in which the temporal order itself carries substantial prognostic information, such as cycle-level degradation trajectories, partial-cycle observations, or sequentially updated health indicators. LSTM networks maintain an internal memory cell and use gating mechanisms to regulate information flow, which mitigates the vanishing gradient problem inherent in basic recurrent networks. At time step t, the key LSTM gates and state updates can be expressed as

$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f), i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i)$$

$$o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o), \widetilde{c}_t = \tanh(W_c x_t + U_c h_{t-1} + b_c)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \widetilde{c}_t, h_t = o_t \odot \tanh(c_t)$$

where x_t is the current input, h_t and c_t are the hidden and cell states, $\sigma(\cdot)$ denotes the sigmoid function, and $\odot$ is the Hadamard product. These gates determine what past information to forget (f_t), what new information to incorporate (i_t and $\widetilde{c}_t$), and how much of the cell state to expose as output (o_t), enabling stable learning of long-term trends in noisy trajectories.

The GRU follows a similar gating philosophy but merges the cell and hidden states and uses only update and reset gates, yielding a more compact recurrent unit with fewer parameters. The GRU update can be formalized as

$$z_t = \sigma(W_z x_t + U_z h_{t-1} + b_z)$$

$$r_t = \sigma(W_r x_t + U_r h_{t-1} + b_r)$$

$$\widetilde{h}_t = \tanh(W_h x_t + U_h(r_t \odot h_{t-1}) + b_h)$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \widetilde{h}_t$$

where z_t controls the interpolation between the previous hidden state and the candidate state $\widetilde{h}_t$, and r_t determines how much past information to incorporate in forming $\widetilde{h}_t$. In practice, GRU is often chosen when a lighter recurrent backbone is preferred, whereas LSTM remains the more established baseline when longer effective memory or stronger modeling flexibility is required.

In the battery prognostics literature, both LSTM- and GRU-based models have been widely used for early-life and partial-trajectory forecasting, where only a fraction of each charging or discharging cycle is observed and the model must infer the future degradation path from incomplete sequential evidence [105–107]. Their attractiveness in this setting stems from the fact that they can exploit temporal structure directly, without requiring the signal to be fully compressed into static hand-crafted descriptors before prediction.

A recurring challenge, however, is cross-condition generalization: recurrent models trained under one chemistry, temperature window, or cycling protocol often degrade when the input distribution shifts. To address this, recent work has combined recurrent backbones with transfer learning, domain adaptation, temporal-convolutional front ends, or mixed recurrent ensembles [16,96,106,108–110]. These studies suggest that much of the reported improvement in LSTM/GRU pipelines arises not from the recurrent cell alone, but from how recurrence is integrated with feature extraction, transfer mechanisms, or robustness-enhancing front ends. Recurrent models are also increasingly embedded in broader probabilistic or hybrid pipelines.

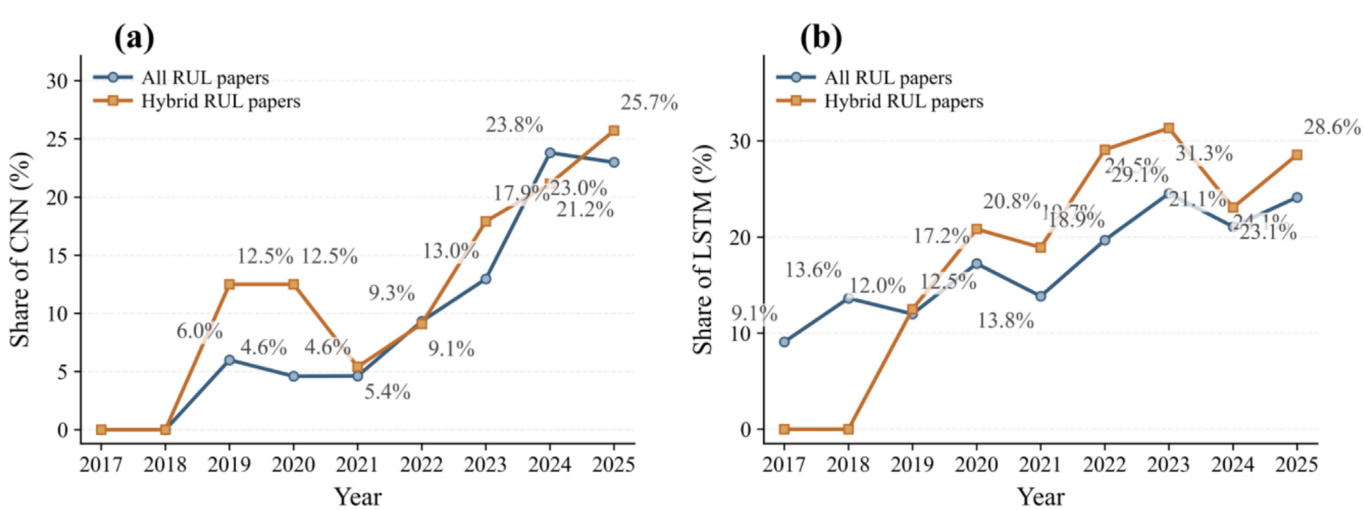

Fig. 10. (a) Trends of CNN adoption; (b) trends of LSTM adoption.

CNN feature extractors combined with BiLSTM and attention can improve signal utilization, while Kalman-filter- or particle-filter-based post-processing can smooth long-horizon trajectories and propagate uncertainty across cycles [111,112]. Bayesian recurrent implementations using Monte-Carlo dropout, as well as sequence-to-sequence formulations that forecast future capacity trajectories rather than a single scalar RUL, further extend this family toward uncertainty-aware and regime-adaptive prognostics [107,113–115]. Signal decomposition strategies such as CEEMDAN are likewise often used to suppress recovery-induced oscillations or high-frequency noise before recurrent prediction [108,116,117].

From a critical perspective, recurrent networks remain among the most credible and widely used deep baselines for battery RUL prediction, especially when the task is naturally sequential and the available data are moderate rather than very large. At the same time, their reported superiority should not be overstated. Performance often depends strongly on window construction, preprocessing, decomposition strategy, and whether the recurrent core is augmented by transfer learning, attention, or hybrid filtering. Moreover, while LSTM and GRU are effective at capturing temporal dependence, they are not always the most efficient choice for very long histories or highly heterogeneous multimodal inputs, where attention-based or graph-based architectures may offer a better inductive bias. Recurrent methods are therefore best viewed as strong sequence-modeling baselines whose value is greatest in cycle-level prognostics, early prediction, and settings where temporal continuity is central to the degradation signal.

As can be seen from Fig. 10(b), although LSTM-based methods accounted for only 9.1% of studies in 2017, in recent years they have consistently maintained a usage share above 20% in RUL prediction, making LSTM one of the core workhorses in this field. Their sustained popularity reflects both methodological maturity and practical flexibility: LSTM- and GRU-based models can serve as standalone predictors, transferable backbones, or temporal modules within larger hybrid prognostic pipelines.

4.3.3. Transformer-based architectures for RUL prediction

Transformer architectures were originally proposed to overcome the limitations of recurrent networks in modeling long sequences, in particular vanishing gradients and strictly sequential computation. In battery RUL prediction, they are most attractive when the task involves relatively long sequential contexts, heterogeneous input channels, or cross-condition variation that may benefit from more flexible dependency modeling than standard recurrent architectures. Instead of propagating information step by step, they rely on a self-attention mechanism, which allows the model to weigh information from different time steps when forming each representation and thus directly relate all positions in a sequence. Formally, given input representations $X = [x_1, \cdots, x_T] \in \mathbb{R}^{T\times d}$, the scaled dot-product self-attention computes attention weights and context as

$$\text{Attention}(Q,K,V) = \text{softmax}\left(\frac{QK^{\top}}{\sqrt{d_k}}\right)V$$

where $Q = XW^Q$, $K = XW^K$, $V = XW^V$ are the query, key, and value matrices obtained via learned projections W^Q, W^K, W^V and d_k is the dimensionality of the keys [118]. This formulation enables the model to capture long-range temporal and cross-channel dependencies without relying on strictly sequential state propagation.

To enrich expressivity, the basic self-attention is extended in multi-head form. With h parallel heads, the multi-head attention output is given by

$$\text{MultiHead}(Q,K,V) = \text{Concat}(\text{head}_1, \cdots, \text{head}_h)W^O$$

$$\text{head}_i = \text{Attention}\left(QW_i^Q, KW_i^K, VW_i^V\right)$$

where each head uses independent projections to capture distinct aspects of the sequence. Position information, which is not inherently encoded by attention, is provided by positional encodings $PE(t) \in \mathbb{R}^d$ added to the input embeddings, enabling the model to distinguish different time steps.

As illustrated in Fig. 9, a Transformer encoder consists of stacked layers, each applying multi-head self-attention, positional encodings, and position-wise feed-forward networks, typically with residual connections and layer normalization [119,120]. When adapted to battery trajectories, the input at each time step is a feature vector summarizing a cycle or sliding window, and the encoder produces a sequence of latent representations that can be pooled or decoded to forecast capacity evolution and remaining life. In practice, this makes Transformer-based models particularly suitable for battery datasets in which the prognostic signal is distributed across relatively long temporal spans or across multiple coupled measurement channels.

Early Transformer variants for Li-ion RUL first denoise the raw capacity or voltage-current sequences and then feed the cleaned series into an encoder to forecast the degradation horizon, achieving competitive accuracy under noisy laboratory data [121,122]. Beyond vanilla encoders, temporal-fusion style designs combine sequence-to-sequence layers with attention over both past and future covariates, which improves multi-horizon capacity and RUL forecasting when operating conditions such as temperature and C-rate vary over time [118]. Multimodal attention networks further fuse cross-channel signals (e.g., voltage, temperature, and internal resistance) and apply temporal attention to focus on informative segments, thereby enhancing robustness under changing regimes [123]. To scale model capacity without incurring quadratic attention costs on all tokens, recent work introduces attention with a mixture-of-experts routing: tokens are dispatched to specialized expert blocks so that only a subset of parameters is activated for each time step, which improves scalability and delivers additional gains on public RUL benchmarks [124]. Taken together, these studies suggest that the main appeal of Transformer-based pipelines in battery RUL prediction lies in their flexibility for modeling long-range, multivariate, and potentially heterogeneous sequential structure, especially when richer telemetry is available than in early benchmark settings.

From a critical perspective, however, the strengths of Transformer architectures should be interpreted with caution. Their reported gains are often most convincing in settings with sufficiently rich inputs, careful preprocessing, and enough data to support attention-based representation learning; they are not automatically superior on the small and highly fragmented datasets that still characterize much of the battery RUL literature. In addition, performance often depends strongly on window design, tokenization strategy, denoising, and whether the Transformer is used as a standalone encoder or as part of a larger multimodal or hybrid pipeline. Transformer-based models are therefore best viewed not as universal replacements for recurrent methods, but as powerful sequence learners whose value is greatest when long-range dependency modeling and heterogeneous signal fusion are central to the prognostic task.

4.3.4. Graph neural networks

GNNs represent degradation structure by casting cycles, voltage segments, sensors, or cells in a pack as nodes in a graph and learning how information propagates along physically meaningful edges. In battery RUL prediction, they are particularly relevant when the prognostic task is not purely sequential, but also depends on structured interactions among cells, signal segments, or sensing channels. As illustrated in Fig. 9, a typical GNN layer takes a node-feature matrix and an adjacency structure, aggregates messages from each node's neighbors through graph convolution or attention, and stacks several such layers before a readout module produces sequence-level or cell-level representations for RUL regression. On this basis, recent studies have shown that dual-attention, conditional, and other spatio-temporal GNN variants can provide strong performance in pack- or segment-level prognostics, especially when graph structure carries information that is difficult to represent with purely sequence-based models [58,125–127].

A central design choice is how to cast battery data into a graph. Common constructions build k-nearest-neighbor or fully connected graphs over voltage segments or cycles, with edge weights defined by dynamic time warping distances, correlations, or related similarity measures; multi-view or nested graphs then fuse several similarity notions to improve robustness under noisy and nonstationary aging [128]. For pack-level problems, nodes typically represent individual cells, edges reflect electrical or thermal coupling, and temporal self-attention on top of graph layers has been shown to improve long-horizon forecasts under fast charging and uneven loading [129]. Graph models developed first for SOH estimation also transfer naturally to RUL by either regressing remaining cycles directly or forecasting the future capacity trajectory and integrating to end-of-life [130]. This means that the practical value of GNNs in battery prognostics depends less on the graph formalism itself than on whether the graph construction genuinely captures degradation-relevant relations, rather than merely re-encoding information that simpler sequence or tabular models could already exploit.

In practice, two implementation patterns dominate: the first combines GCN or GAT blocks with a separate temporal encoder such as a TCN, LSTM, or Transformer; the second uses unified spatio-temporal GNNs in which time is encoded in edges or through temporal attention [125,126]. To mitigate oversmoothing, topology shift, and gradient instability over long aging horizons, recent studies explore curriculum over window length, edge or neighborhood dropout, dynamic and pruned graphs, and adaptive loss designs [128,129,131]. Taken together, the current literature suggests that GNNs are most attractive when informative interactions between segments, cycles, or cells play a central role in the prognostic target, particularly in module- and pack-level settings where electrical or thermal coupling is non-negligible.

From a critical perspective, however, the reported advantages of GNN-based RUL models should be interpreted with caution. Their effectiveness is often highly sensitive to graph construction choices, including node definition, edge semantics, similarity metrics, and the extent to which temporal information is incorporated explicitly. As a result, reported performance gains may reflect not only the GNN architecture itself, but also the quality of the underlying graph design. Moreover, when the task is dominated by single-cell temporal evolution rather than relational structure, GNNs may offer limited benefit relative to simpler recurrent or attention-based sequence models while incurring greater computational and implementation complexity. GNNs are therefore best viewed as structurally informed predictors whose value is greatest when interaction topology is a genuine part of the degradation problem, rather than as universal replacements for other deep architectures.

4.4. Ensemble data-driven methods

In recent years, an increasing number of studies have adopted multi-model ensemble data-driven strategies for battery RUL prediction in order to exploit the complementary strengths of different learning algorithms. Here, "ensemble" refers specifically to combinations of multiple data-driven components, whereas "hybrid" methods incorporate both data-driven and model-based elements [132,133]. This distinction is important because the two paradigms address different limitations: the former seeks stronger predictive performance through algorithmic complementarity, while the latter introduces physical structure or mechanistic priors into the prognostic pipeline. As illustrated in Fig. 11, representative ensemble approaches combine CNNs, which are effective at extracting local temporal or spatial patterns, with LSTM-based sequence modeling to capture both short-term signal structure and longer-term degradation dynamics [134–136]. Reported results suggest that such ensemble data-driven methods can outperform standalone predictors, particularly under noisy measurements, irregular cycling protocols, or strongly nonlinear degradation trajectories [137,138].

Ensemble data-driven pipelines increasingly combine kernel-based regression, sequence modeling, and signal preprocessing modules in order to capture nonlinear mappings and temporal dependencies simultaneously. In this spirit, SVR has been integrated with LSTM under an optimization-constrained formulation, where the standard deviation is used as a criterion and the resulting constrained multivariate objective is refined via a sequential quadratic programming routine [139]. Related workflows further combine signal decomposition and multi-model regression by applying variational mode decomposition (VMD) to accelerate vibration-signal processing and feeding the decomposed instantaneous-frequency components into an LSTM-SVR model for damage-trend prediction [140]. Beyond SVR-RNN stacks, CNN-LSTM fusion has been developed as an organic spatiotemporal learner rather than as a simple CNN feature extractor; in particular, when single-timestamp inputs are used, padding signals within the same training batch can adversely affect predictive performance and therefore requires careful handling [141]. Convolution has also been embedded directly into both the input-to-state and state-to-state transitions of LSTM, enabling joint modeling of time-frequency structures and temporal dynamics while retaining the benefits of gated memory [142]. For probabilistic RUL estimation, health factors extracted from charging curves can be screened via the maximum information coefficient and then used to build aging and RUL predictors with GPR [91], thereby providing calibrated uncertainty alongside point predictions through its Bayesian formulation [143]. More generally, recent studies increasingly integrate three or more data-driven components to improve robustness from multiple angles and to better accommodate diverse practical requirements in RUL prediction [94,144,145].

From a critical perspective, the value of ensemble data-driven methods should not be attributed merely to the fact that multiple models are combined. Their reported gains often depend strongly on how the constituent modules are organized, what information each component contributes, and whether the resulting pipeline genuinely captures complementary aspects of degradation rather than adding architectural complexity without clear functional benefit. In practice, these methods are most convincing when a single model is unlikely to handle local feature extraction, temporal dependency modeling, uncertainty awareness, and noise suppression equally well. At the same time, ensemble pipelines often incur higher computational cost, more difficult tuning, and reduced interpretability relative to simpler standalone predictors [146–149]. They should therefore be viewed not as universally superior solutions, but as flexible data-driven combinations whose effectiveness depends on the match between model complementarity, data characteristics, and the target prognostic scenario. A different route to improving robustness and credibility is to inject physical structure or mechanistic constraints into the learning process, which motivates the hybrid and physics-informed methods discussed in subsection 4.5.

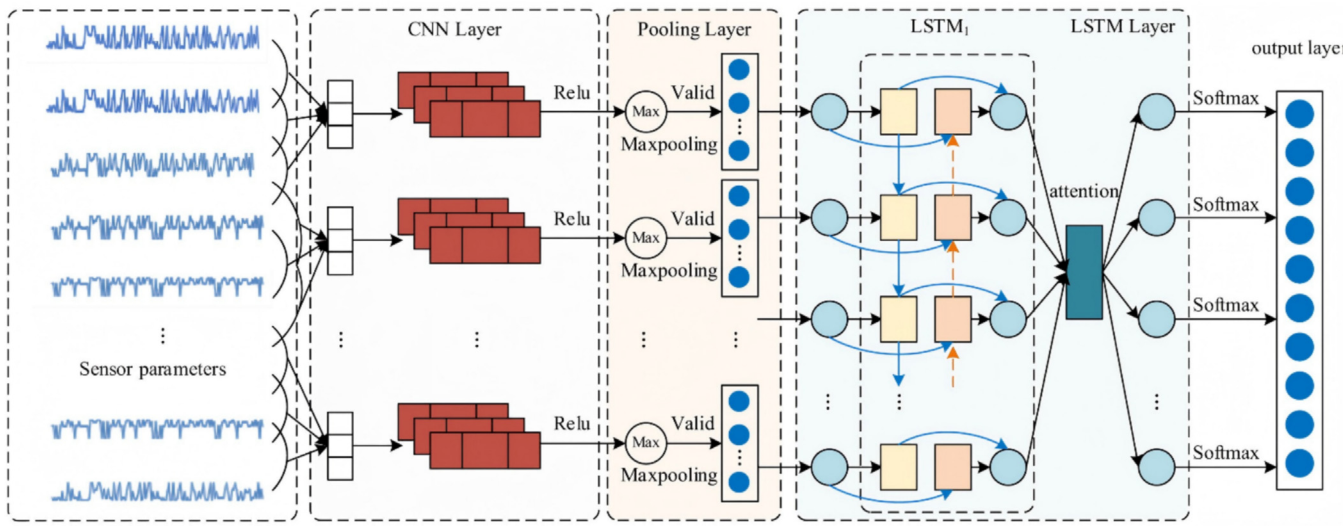

Fig. 11. An example of ensemble data-driven RUL prediction framework: LSTM+CNN+Attention. Reproduced with permission from Ref. [136]. Copyright 2024, The Authors.

4.5. Hybrid and physics-informed methods

Hybrid methods refer to RUL prediction approaches that integrate both model-based and data-driven techniques. As is shown in Fig. 12 and Table 6, compared with purely data-driven approaches, the incorporation of physics-based structures, mechanistic priors, or physically constrained state representations can improve physical consistency and, in some settings, enhance interpretability and extrapolation credibility. Recent survey papers on lithium-ion battery PHM consistently highlight such hybrid strategies as a key research direction, typically grouping them into (i) physics-based models augmented with learned correction terms, (ii) state-space filters whose transition or observation models are partially data-driven, and (iii) tightly coupled physics-informed neural networks (PINNs) that encode governing equations as soft constraints during training.

Recent progress has made these three directions more differentiated in both methodology and application scope. Physics-guided correction hybrids are increasingly used when simplified electrochemical or equivalent-circuit models can provide a physically meaningful baseline, while the learning component compensates for unmodelled dynamics, operating-condition variation, or capacity-regeneration behavior [150,151]. State-space filtering hybrids have become especially relevant for online prognostics because they combine recursive state updating, uncertainty propagation, and noise robustness within a unified estimation framework, which is attractive for deployment-oriented battery health management [152,153]. Physics-informed neural hybrids, in turn, have emerged as one of the fastest-growing branches, especially in studies that seek to improve data efficiency, preserve physically plausible degradation trajectories, or regularize learning under distribution shift [154,155].

This recent evolution also indicates that the main value of hybrid methods is not simply higher point accuracy. Rather, their advantages are most meaningful when the prognostic task involves small or heterogeneous datasets, strong operating-condition shift, explicit uncertainty requirements, or a need to maintain consistency with known degradation mechanisms. For example, recent physics-informed machine learning frameworks have shown that early-stage lifecycle data can be combined with mechanistic priors to support RUL prediction under limited run-to-failure observations [151], while recent PINN-based battery health frameworks have further demonstrated how physical constraints can stabilize learning and improve robustness under varying operating conditions [154,155]. At the same time, the performance of hybrid approaches remains strongly dependent on how physical knowledge is injected, whether the adopted prior is identifiable and compatible with the available data granularity, and how the physics-based and data-driven objectives are balanced during training.

As battery applications move toward safety-critical and highly variable real-world scenarios, these hybrid RUL frameworks are therefore expected to play an increasingly central role in bridging the gap between laboratory models and field data and in enabling trustworthy deployment in practice. Our bibliographic analysis indicates that the role of hybrid methods has become more visible in recent RUL studies, especially in work addressing limited data, operating-condition shift, uncertainty-aware prediction, and physics-consistent degradation modeling. However, this trend is better interpreted as a task-driven methodological shift rather than as a simple ranking of model popularity. The representative categories summarized in Table 7 therefore reflect not only conceptual differences, but also the main ways in which recent hybrid battery prognostics research has progressed beyond purely data-driven prediction.

5. Challenges

While recent machine-learning approaches have markedly improved the efficiency and accuracy of lithium-ion battery RUL prediction, substantial challenges remain. At a high level, these open issues can be grouped into three dimensions as is shown in Fig. 13: (i) data-related challenges, including limited coverage, noise and scenario gaps; (ii) methodological challenges, concerning feature representations, model design and physics-data fusion; and (iii) application-level challenges, such as deployment on resource-constrained BMS hardware, robustness under real-world operating conditions, and the interpretability of prognostic outputs. The following subsections discuss these three dimensions in turn.

5.1. Data-related challenges for RUL prediction

Public battery datasets have catalyzed progress in data-driven RUL research, yet several structural limitations remain that can bias results and complicate deployment.

Limited coverage and condition shift. Most publicly available datasets are collected under tightly controlled laboratory condi-

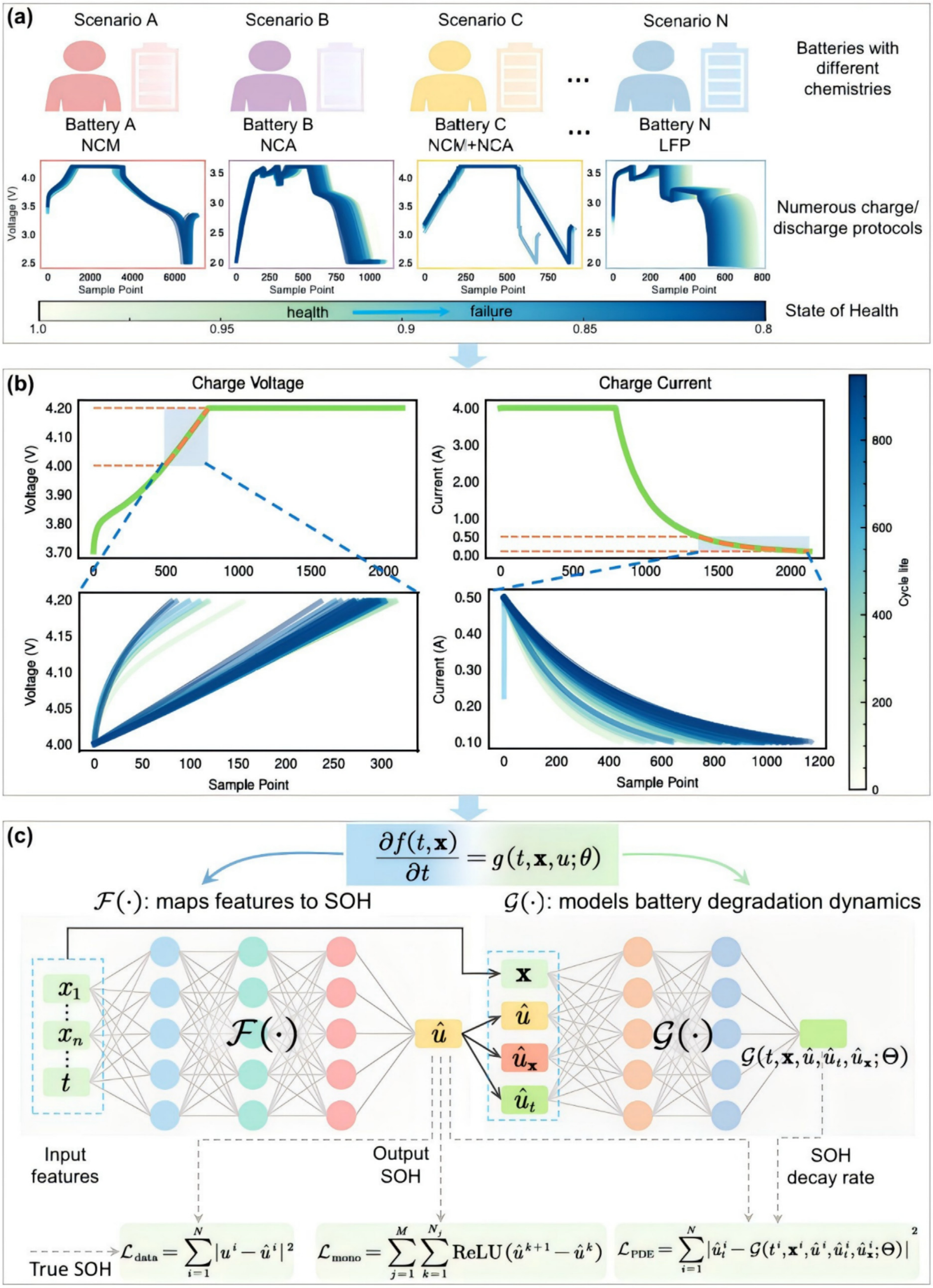

Fig. 12. An example of PINN. (a) The lithium-ion batteries may have different chemistries. (b) An illustration of the selected data for feature extraction. (c) The structure of the proposed PINN. Reproduced with permission from Ref. [25]. Copyright 2024, Elsevier.

Table 6
Main categories of hybrid model-based and data-driven methods for Li-ion battery RUL prediction.

Category	Key idea	Refs.
Physics-guided correction hybrids	A physics-based model provides a baseline RUL estimate, and machine-learning components learn residuals or physics-informed health indicators to correct for unmodelled dynamics and changing operating conditions	Zhang et al. [229]; Li et al. [230]
State-space filtering hybrids	Degradation is formulated in a state-space model, with some transition or observation terms learned from data and fused through Bayesian filtering to yield probabilistic RUL estimates under noisy measurements	Cadini et al. [152]; Pan et al. [153]
Physics-informed neural hybrids	Neural networks are trained under explicit physical constraints so that learned representations and RUL trajectories remain consistent with governing laws and degradation mechanisms	Wen et al. [231]; Ma et al. [176]; Deng et al. [151]

tions with fixed temperatures and pre-defined load profiles, which only imperfectly represent real-world operating duty cycles. Consequently, models trained on such data often exhibit degraded performance when deployed under shifts in battery chemistry, temperature regime, or load profile. Many widely used datasets concentrate on a single chemistry or very small cohorts, such as the LFP-graphite 18,650 cells in the MIT dataset or the eight LCO pouch cells in the Oxford OBD-1 study, and even datasets based on standardized drive cycles like UDDS remain limited in both size and condition diversity [14,41]. These constraints restrict what can be concluded about cell-to-cell variability and transferability, and they leave open important questions regarding model robustness under broader field conditions [45].

In practice, this means that many reported gains are still established under relatively narrow in-distribution settings, rather than under the chemistry, protocol, or temperature shifts that arise in deployment. As a result, cross-condition, cross-chemistry, and cross-dataset validation should be treated as a more informative complement to within-dataset testing, and future dataset development should place greater emphasis on metadata standardization and condition diversity to support fairer evaluation and more reliable transfer.

EOL and metadata inconsistency. EOL criteria vary across datasets: the MIT fast-charging set defines EOL at 80% of nominal capacity, whereas NASA's aging tests terminate at 30% capacity fade—so nominally identical "RUL" targets are not directly comparable without harmonization [14,39]. Historical releases also differ in file structures, metadata and annotation granularity, raising the cost of cross-study benchmarking and automated preprocessing. Standardization efforts such as Battery Archive help normalize fields and provide study summaries, but coverage is still maturing [42].

Beyond a terminological mismatch, inconsistent EOL definitions change the supervised target itself: a model trained to predict the remaining cycles to an 80% capacity threshold is solving a materially different task from one trained to a 70% threshold or a 30% fade criterion. This difference alters the effective prediction horizon and weakens the comparability of reported MAE or RMSE across studies unless the failure threshold, capacity normalization rule, and evaluation window are explicitly aligned. The problem is further compounded by heterogeneous metadata conventions across datasets, which have been identified as a central barrier to large-scale battery data science and cross-source reuse [156,157].

Sparse diagnostics. Continuous voltage-current-temperature streams are common, but regularized diagnostics are often missing or performed only sparsely, limiting mechanistic interpretability and multi-modal supervision. In many campaigns these tests are run on a small subset of cells or at a few fixed life stages, which makes it difficult to track how internal resistance, diffusion limitations and side reactions evolve over time. As a result, hybrid and physics-informed models that rely on such signals must either operate with weak supervision or fall back to proxy features extracted from cycling data alone. Recent relaxation-focused datasets begin to address this gap [158].

Beyond limiting mechanistic insight, sparse diagnostics also make it difficult to compare multi-modal methods fairly across studies, because models that exploit EIS, HPPC, or relaxation information are often trained and evaluated on data conditions that are not consistently available in conventional benchmark datasets. A practical direction for future dataset design is therefore to include a minimum standardized cadence of diagnostic measurements, even if only at several representative aging stages, so that physically grounded features can be incorporated more systematically and cross-study evaluation of hybrid models becomes more reproducible.

The above discussion has focused on the limitations of widely used public battery datasets. In what follows, the challenges that arise when researchers construct their own datasets will be analyzed in detail.

Key barriers in dataset acquisition. A direct remedy for the limitations of public datasets is to generate proprietary run-to-failure data. In practice, however, collecting end-of-life trajectories at scale is exceptionally resource-intensive: a single experiment often spans months to years, even before considering the parallel cohorts required for statistical coverage [159,160]. Moreover, systematic exploration of protocol space without automation can consume hundreds of days, and several open datasets report cells aged for well over a year prior to release [14,161,162]. These time horizons and throughput constraints place such campaigns beyond the realistic means of many laboratories.

Beyond limiting data volume, this acquisition bottleneck also shapes the kinds of scientific claims that can be made: small proprietary cohorts may be sufficient for proof-of-concept modeling, but they rarely support rigorous assessment of variability across cells, protocols, and environmental conditions. As a result, methodological advances are often validated on narrow experimental domains, which increases the risk that apparent performance gains reflect dataset convenience rather than robust prognostic capability. A more practical path forward is therefore not only to accelerate experimentation through automation and closed-loop design, but also to prioritize interoperable data formats and cross-institutional data accumulation so that statistical diversity can improve without requiring every laboratory to bear the full cost of large-scale aging campaigns.

Experimental design, sunk-cost bias, and domain shift. Dataset design governs both the learning target and the empirical data distribution, because experimental choices such as voltage limits, charge protocols, temperature windows, rest periods, and diagnostic cadence directly affect the observed cycle life and its variability. The MIT fast-charging dataset provides a clear example: charging policy alone can induce orders-of-magnitude differences in lifetime. This, in turn, underscores the need for efficient experimental design to avoid costly but weakly informative aging campaigns [14,161]. In contrast, the NASA PCoE aging dataset fixes both end-of-life criteria and cycling protocols, for example by imposing specific voltage cutoffs and defining end of life at a 30% capacity loss; this standardization simplifies labeling yet may limit transferability to other usage regimes [40]. Because life testing is slow and expensive, sub-optimal protocol choices can lock in substantial sunk costs when the resulting data generalize poorly. Recent community initiatives, such as the Battery Data Genome project, there-

Table 7
Representative ensemble and hybrid data-driven methods for Li-ion battery RUL prediction.

Refs.	Year	Methods	Method characteristics	Datasets	Results
Khumprom et al. [232]	2019	DNN	End-to-end nonlinear regression with minimal feature engineering	NASA	RMSE = 5 cycles
Kong et al. [233]	2019	HI + CNN + LSTM	Local feature extraction combined with temporal degradation modeling	NASA	RMSE = 16.127 cycles
Zraibi et al. [137]	2021	CNN + LSTM + DNN	Hierarchical feature learning with temporal dependency modeling	NASA, CALCE	RMSE = 0.0043 cycles; R^2 = 0.99935
Ren et al. [234]	2021	Autoencoder + CNN + LSTM	Unsupervised feature compression followed by spatio-temporal learning	NASA	NRMSE = 5.03%
Jafari et al. [86]	2022	PF + KF + XGBoost	Probabilistic filtering combined with tree-based regression	NASA	NRMSE = 0.0179%
Li et al. [138]	2023	TCN + GRU + DNN + Attention	Multi-scale temporal modeling with attention mechanism	NASA, CALCE	NRMSE = 2.407%
Zhao et al. [235]	2024	Autoencoder + Transformer	Global dependency modeling on compact latent representations	MIT	RMSE = 16 cycles
Shi et al. [236]	2024	SSA + LSTM	Meta-heuristic optimization of recurrent network parameters	NASA	NRMSE = 2.24%
Safavi et al. [237]	2024	COM + CNN + XGBoost	Optimized hybrid of deep feature extraction and ensemble regression	NASA	RMSE = 106 cycles; MAPE = 7.5%
Ma et al. [238]	2025	XGBoost + LSTM	Stacked ensemble integrating tree models and temporal learners	NASA	RMSE = 0.0109 cycles; MAPE = 0.7165%
Fan et al. [27]	2025	CEEMDAN + SVR + LSTM + SSA	Signal decomposition with optimized hybrid regression and temporal modeling	NASA	RMSE = 0.0066; MAE = 0.0046
Duan et al. [78]	2023	IWOA + OS-ELM + PF	Online learning with adaptive forgetting and probabilistic state filtering	NASA	NRMSE = 4.167%
Zhang et al. [239]	2022	Attention + BiLSTM	Attention-enhanced bidirectional temporal dependency modeling	NASA, CALCE	NRMSE = 1.73%; R^2 = 0.9678
Gao et al. [240]	2022	CNN + BiLSTM	Spatial feature extraction combined with bidirectional sequence learning	NASA	MAE = 0.005%; RMSE = 0.007716
Hu et al. [241]	2022	Wavelet Denoising + Transformer	Noise suppression with global dependency modeling via self-attention	CALCE, AQNU	MAE = 1.81%; RMSE = 2.19%
Chen et al. [242]	2023	CNN + LSTM	Feature relevance enhancement with spatio-temporal fusion learning	NASA	MAPE = 0.36%
Qiao et al. [243]	2020	EMD + DNN + LSTM	Decomposition-based feature extraction with deep temporal regression	NASA	RMSE = 0.0025
Li et al. [244]	2023	GPR + WOA + Autoencoder	Probabilistic regression with feature compression and meta-heuristic tuning	NASA	RMSE = 1.81%; MAE = 1.61%
Qi et al. [245]	2025	EEMD + GPR + LSTM	Decomposition with probabilistic regression and temporal modeling	NASA	RMSE = 0.0566; R^2 = 0.9612; MAE = 0.0455
Liu et al. [246]	2024	CEEMDAN + PSO + GRU	Decomposition with *meta*-heuristic tuning for small-sample sequence learning	NASA, CALCE	MAE = 0.00632; RMSE = 0.00901
Zhang et al. [81]	2023	ELM + RBF + ISSA	Fast shallow learner with kernel mapping and optimized hyperparameters	NASA	RMSE = 0.0011; MAPE = 0.07%
Sun et al. [247]	2024	SVD + SVAE	Compact feature extraction with uncertainty-aware regression	NASA	RMSE = 0.0116; R^2 = 0.9921
Chen et al. [124]	2024	MoE + Attention	Expert routing to scale attention while improving specialization	NASA, CALCE	RMSE = 0.0872; MAE = 0.0815
Ma et al. [248]	2024	GNN + Autoencoder + Transformer	Graph representation learning with global temporal dependency modeling	NASA	MAE = 0.0993; RMSE = 0.0044
Xu et al. [249]	2023	DNN	Ensemble-style deep regression for early-life forecasting	NASA	RMSE = 114.05 cycles

fore advocate standardized metadata and open sharing practices to improve comparability and reuse [163].

Even large laboratory datasets are typically collected under tightly controlled conditions, whereas employed batteries experience heterogeneous and evolving duty cycles. This laboratory-field mismatch introduces domain shift, so strong in-laboratory performance may not translate into robust deployment [36]. Future work should therefore emphasize deployment-oriented benchmarks and domain-generalizable prognostic models.

Proprietary datasets and barriers to reproducibility. Because run-to-failure data acquisition is inherently costly and time-consuming, some studies choose not to release the datasets underlying their results, treating the resulting records as proprietary assets. Although the substantial effort required to generate high-quality battery aging data deserves full recognition, limited data

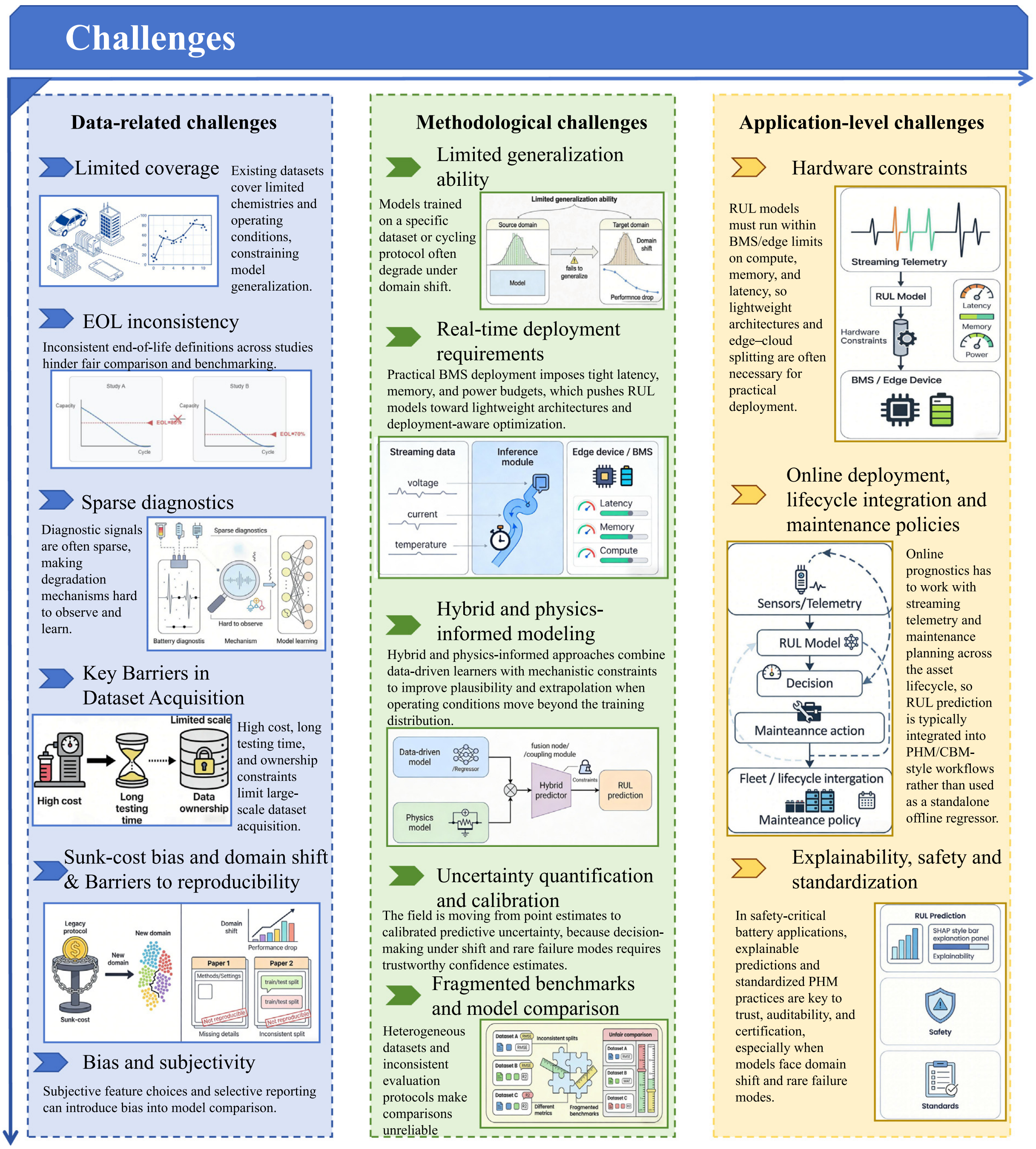

Fig. 13. Challenges of data-driven RUL prediction.

sharing across the field constrains reproducibility, impedes fair benchmarking, and slows cumulative progress in RUL prognostics. Community initiatives that promote open, well-annotated datasets and interoperable metadata, such as the Battery Data Genome project, aim to alleviate these constraints by aligning battery data practices with the widely adopted principles of findability, accessibility, interoperability, and reusability [163,164]. In this sense, broader adherence to these practices should be viewed not simply as a matter of professional courtesy, but as an essential means of accelerating collective learning and reducing duplicated effort [163,164].

Beyond preventing exact replication, restricted data access also obscures preprocessing choices, end-of-life labeling rules, and exclusion criteria, making it difficult to determine whether reported gains arise from superior modeling or from dataset-specific handling. In practice, this weakens cross-study comparability and slows the emergence of robust community benchmarks. A realistic way forward is therefore not only broader data release, but also the routine publication of machine-readable metadata, preprocessing scripts, and minimally reusable benchmark splits, so that even partially shared datasets can support more transparent and cumulative progress [157,165].

Bias and subjectivity in public datasets. Even when publicly released, battery datasets inevitably reflect subjective choices in experimental design, curation, and preprocessing. Decisions regarding sample exclusion, smoothing or denoising, resampling, relabeling, and other cleaning operations can alter both the effective input distribution and the target definition seen by the model, thereby influencing not only training outcomes but also the interpretation of reported prediction errors. In practice, such flexibility can make apparent performance gains partly contingent on dataset-specific handling rather than on genuinely superior prognostic capability, which weakens cross-study comparability and reduces the reliability of benchmark-based conclusions. Evidence from the broader machine-learning literature shows that benchmark datasets often contain non-negligible label errors and that undisclosed analytic flexibility can materially distort empirical evaluation [166–169].

For battery RUL research, this implies that transparent reporting of preprocessing decisions, label construction rules, exclusion criteria, and benchmark splits is not optional but essential. Initiatives emphasizing interoperable metadata and explicit provenance, including the FAIR principles and the Battery Data Genome, therefore provide not only a framework for openness, but also a practical basis for more reproducible model training, fairer accuracy evaluation, and more credible cross-study comparison [164,170].

5.2. Methodological challenges in ML-based RUL prediction

Even with access to high-quality corpora, battery RUL predictors still face several practical limitations. First, generalization remains fragile: models trained under a narrow set of chemistries, temperatures, or cycling protocols often exhibit sizable errors once the operating distribution shifts, undermining reliability in real deployments [36,171]. Second, deployability is constrained by runtime and memory budgets in embedded BMS/CPS: many state-of-the-art deep models are computationally heavy, and their predictions must be accompanied by calibrated uncertainty to support safety-critical decisions [172,173]. Third, there is no single dominant method: comparative reviews continue to report dataset- and horizon-dependent rankings, with hybrids frequently outperforming standalones on one benchmark while lagging on another [171,174,175]. Finally, promising directions—such as physics-informed or hybrid learning and explicit domain adaptation/generalization, i.e., adapting models to related new operating domains or improving robustness to previously unseen ones—are active areas of research rather than solved engineering recipes [176,177].

Limited generalization ability. Generalization refers to a model's ability to remain accurate on data collected under operating conditions, chemistries, or protocols that differ from those seen during training [178]. In battery RUL prognostics, this property is essential because real deployments routinely involve shifts in temperature, C-rate, rest time, and even end-of-life definitions, so a method that performs well on a single laboratory dataset may still have limited practical value [36]. This challenge is reinforced by the strong sensitivity of cycle life to charging protocol and policy choices, as demonstrated by the MIT fast-charging studies [14,161].

Accordingly, many purely data-driven RUL pipelines exhibit degraded performance under distribution shift, revealing fragile out-of-distribution behavior and weak transfer across chemistry, temperature, and usage profiles [36]. To address this, recent studies have increasingly explored transfer learning and domain adaptation, including self-attention-based unsupervised adaptation, PSO-assisted deep adaptation, fine-tuned transformer backbones, cross-chemistry feature alignment, and EIS-informed transfer [179–183]. Taken together, these results suggest that future RUL research should prioritize robust generalization across domains, rather than optimizing only for in-distribution accuracy.

Real-time deployment requirements. In operational settings such as electric vehicles, RUL predictors must deliver not only accurate but also low-latency estimates, since battery management systems (BMS) require continuously updated predictions to support control and safety decisions in real time [36,184]. However, many state-of-the-art models remain computationally intensive and memory-demanding, which limits their deployment on embedded BMS hardware with tightly constrained CPU, RAM, and power budgets [184,185].

To narrow this gap between laboratory prototypes and practical deployment, two complementary directions are emerging. One is the use of online filtering and hybrid schemes, such as neural networks coupled with adaptive or unscented Kalman filters and particle filters, to enable fast streaming updates with uncertainty awareness [186]. The other is edge-oriented model compression and quantization, which reduces memory and latency while preserving acceptable accuracy for embedded inference [187]. Looking ahead, practical EV RUL prediction will require deployment-aware methods that jointly optimize accuracy, latency, and uncertainty calibration under hardware constraints, rather than accuracy alone [36,185].

Hybrid and physics-informed modeling. A challenge lies in how to systematically fuse domain knowledge with flexible function approximators. Physics-informed or hybrid architectures promise better extrapolation and interpretability, but in practice they introduce additional design and optimization difficulties. For example, coupling equivalent-circuit or reduced-order electrochemical models with neural networks requires choices about which states or residuals to learn, how to balance data-fit and physics-consistency losses, and how to handle stiff dynamics and multiple time scales [36,176]. These models often have larger parameter spaces and non-convex multi-objective loss landscapes, making training sensitive to initialization, scaling and hyperparameters. Ensuring numerical stability, identifiability of physical parameters, and robustness of the learned corrections across operating regimes remains an open research problem rather than a solved engineering routine.

Beyond posing optimization challenges, these design choices have direct implications for practical research. Reported gains may partly depend on how physical constraints are selected, weighted, and enforced, rather than reflecting genuinely superior extrapolation across chemistries or operating regimes. In addition, physical parameters or latent states learned under one protocol are not always identifiable or transferable under another, which weakens both cross-study comparability and confidence in deployment. Future work should therefore place greater emphasis on benchmark protocols that explicitly report physics-loss design, parameter identifiability, and cross-condition validation, while developing hybrid frameworks that preserve physical consistency without sacrificing optimization stability.

Uncertainty quantification and calibration. Another limitation concerns the treatment of uncertainty in RUL prediction. Most legacy pipelines output point estimates only, even though battery health trajectories are affected by substantial aleatoric variability and epistemic uncertainty like limited data in new conditions [172,173]. Recent work explores Bayesian deep learning, ensembles and quantile regression to provide predictive distributions or intervals, but these methods can be computationally demanding and their calibration quality is rarely reported in a standardized way. In particular, long-horizon RUL forecasts tend to be overconfident, and coverage of prediction intervals can degrade non-uniformly across different degradation regimes. Developing lightweight, well-calibrated uncertainty estimation methods that remain reliable under distribution shift is therefore a key methodological gap.

Beyond methodological incompleteness, insufficiently calibrated uncertainty has direct practical consequences for research and deployment. Overconfident long-horizon forecasts can make model accuracy appear stronger than it is under distribution shift, while poorly reported calibration weakens the comparability of probabilistic methods across studies. Future work should therefore move beyond reporting interval width alone and adopt standardized calibration diagnostics, such as coverage error and reliability analysis, together with cross-condition evaluation, so that uncertainty estimates can be judged not only by sharpness but also by trustworthiness under realistic operating variability.

Fragmented benchmarks and model comparison. Algorithmic progress is further hindered by fragmented benchmarks and inconsistent evaluation protocols. Comparative studies repeatedly show that no single architecture dominates universally; the relative performance of SVR, tree ensembles, CNNs, RNNs, and hybrid models depends strongly on the dataset, prediction horizon, target definition, preprocessing pipeline, and error metric [171,174,175]. Many studies still rely on bespoke train-test splits, ad hoc cell filtering, manually engineered feature sets, and non-unified RUL conventions, often without releasing code or preprocessing scripts. Consequently, reported gains cannot always be attributed unambiguously to genuine modeling advances, but may partly reflect favorable experimental settings. Future progress therefore requires benchmark suites with standardized splits, target definitions, metrics, and baseline implementations, together with transparent preprocessing and open evaluation pipelines, to support fairer and more reproducible comparison in battery RUL prediction.

5.3. Application-level challenges: from models to deployable PHM solutions

Sensing, instrumentation, and BMS deployment constraints. At the application level, a primary challenge is the migration of promising RUL models into production BMS operating under tight sensing and hardware constraints. In real battery packs, prognostics must share limited microcontroller resources with core BMS functions such as protection, state estimation, and balancing, which severely restrict the CPU time, memory, and storage available for RUL inference. In addition, sensing configurations are often sparse and heterogeneous: many systems provide only pack-level voltage, a limited number of temperature probes, and a single current sensor, with occasional dropouts, recalibrations, and firmware changes. Models developed on rich laboratory datasets therefore often require compression, simplification, or partial cloud support, and must be redesigned to tolerate low-rate, noisy, and potentially biased measurements. Heterogeneity across vehicle platforms and energy-storage products further complicates the reuse of a single trained model without careful adaptation and validation.

These constraints imply that strong offline accuracy alone is not sufficient to establish practical utility. A model that performs well on laboratory benchmarks but cannot satisfy latency, memory, or sensing limitations on embedded BMS hardware has limited deployment value. Future work should therefore place greater emphasis on deployment-oriented evaluation protocols that report not only prediction error, but also inference latency, memory footprint, and robustness to sparse or noisy sensing. In parallel, closer co-design of sensing, algorithms, and hardware will be needed so that prognostic models are developed with realistic BMS constraints in mind, rather than adapted to them only after laboratory validation.

Online deployment, lifecycle integration and maintenance policies. Beyond raw computation, RUL prediction must be embedded into the full asset lifecycle and the operational decision-making process. Most offline studies evaluate performance using error metrics such as RMSE or MAPE on held-out trajectories, but in practical deployments prognostics are used to trigger maintenance, derating or warranty actions under evolving usage patterns. This requires models that can be updated online or periodically recalibrated as operating conditions, user behaviour and degradation mechanisms drift away from the training distribution, ideally leveraging fleet-level data while preserving privacy and robustness. It also calls for RUL outputs that are accompanied by uncertainty information and stable behaviour near end-of-life thresholds, so that operators can define reliable safety margins and avoid both overly conservative early retirements and risky late interventions. Designing PHM functions that interact consistently with existing diagnostic trouble codes, scheduling logic and maintenance policies remains an open systems-engineering problem.

A further implication is that offline predictive accuracy does not translate directly into operational value unless RUL outputs are evaluated in the context of downstream decisions. In practice, maintenance triggers, derating rules, and warranty thresholds depend not only on point accuracy, but also on update frequency, uncertainty quality, and the stability of predictions under changing usage patterns. Future work should therefore move beyond trajectory-level error metrics alone and adopt decision-aware evaluation protocols that quantify the value of prognostic outputs for maintenance timing, safety margins, and lifecycle management. In parallel, greater attention should be given to online recalibration, fleet-level learning, and privacy-preserving update mechanisms so that prognostic models can remain useful as batteries age under heterogeneous real-world conditions.

Explainability, safety and standardization of RUL predictions. Finally, explainability, safety, and standardization remain major barriers to the large-scale deployment of ML-based RUL models. Many high-performing deep architectures still behave as black boxes, making it difficult for engineers, regulators, and operators to determine which signals drive the predictions or to diagnose failure modes when the model is wrong. For safety-critical applications, prognostic functions must also behave predictably under sensor faults, communication errors, and out-of-distribution inputs, while remaining compatible with broader functional-safety and cybersecurity requirements. At the same time, the lack of widely accepted benchmark datasets, deployed-setting evaluation protocols, and reporting standards continues to hinder fair comparison and practical certification of methods.

These issues have direct consequences for deployment: limited interpretability weakens trust in maintenance or derating decisions, insufficient safety validation constrains certification, and non-standardized reporting makes it difficult to judge whether performance differences reflect genuine modeling advances or merely different evaluation conditions. Future work should therefore move toward more transparent prognostic frameworks, including explicit reporting of input signals, preprocessing steps, failure thresholds, uncertainty behavior, and fault tolerance under abnormal conditions. In parallel, benchmark development should extend beyond offline accuracy alone to include explainability, calibration, robustness, and safety-relevant stress testing, so that ML-based RUL models can be assessed not only for predictive performance, but also for deployability and trustworthiness.

6. Future directions

In this section, we provide a task-oriented outlook on future research directions for ML-based Li-ion battery RUL prediction. Instead of relying on a separate method-popularity matrix, we synthesize the bibliographic patterns identified above together with the methodological and application-level limitations discussed in the preceding sections. Overall, recent studies indicate a clear movement from isolated accuracy-driven predictors toward more robust, transferable, uncertainty-aware, and physics-consistent prognostic frameworks. This trend motivates the following discussion of several prospective directions, including hybrid and physics-guided learning, privacy-preserving collaboration, uncertainty-aware prognostics, self-supervised and semi-supervised learning, and reinforcement-learning-assisted battery management.

Hybrid approaches: combining complementary strengths. Future progress in battery RUL prediction will likely depend less on proposing yet another standalone architecture than on designing hybrid frameworks that integrate complementary inductive biases in a principled and scalable manner. Rather than combining models in an ad hoc fashion, the central challenge is to determine how temporal encoders, lightweight regressors, and physics-based components should be coupled so that each contributes information that the others cannot reliably provide on their own [36,175]. In this regard, sequence models remain valuable for representing degradation trajectories, margin-based regressors remain attractive in small-sample and resource-constrained settings, and electrochemical or reduced-order models provide physical structure that can improve interpretability and extrapolation credibility [61,109,188]. A particularly promising direction is physics-informed or physics-regularized learning, in which governing constraints, conservation relations, or mechanistic priors are embedded into the estimator so that predictions remain physically plausible under operating- condition shift while retaining the flexibility of modern machine learning [188,189]. Going forward, the field needs hybrid pipelines that are not only more accurate, but also easier to calibrate, computationally tractable for deployment, and robust across chemistries, cycling protocols, and sensing configurations. This, in turn, calls for better-principled hybrid architectures in which model complementarity, physical consistency, and deployment constraints are considered jointly rather than optimized in isolation.

Privacy-preserving and collaborative learning. Direct data sharing across research groups is often constrained by the high cost and long duration of battery aging experiments, the institution-specific nature of testing protocols, concerns over data ownership, publication priority, and the continuing research value of hard-won datasets. Against this background, privacy-preserving collaborative learning has attracted increasing attention, with FL emerging as a representative framework for collaborative model development without centralizing raw data. In a typical FL workflow, each participant trains a local model and transmits only model updates to a coordinating server, which aggregates them (e.g., via FedAvg) and returns an updated global model for the next round. In this way, FL preserves data locality while still allowing the learned model to benefit from cross-site diversity [190,191]. To further reduce information leakage from transmitted updates, secure aggregation protocols can ensure that the server recovers only the aggregate of client updates rather than any individual contribution [192]. Privacy protection can also be strengthened at the algorithmic level. Differential privacy (DP) limits what can be inferred about any individual record, and DP-SGD makes this practical for modern deep models through gradient clipping and calibrated noise injection [193,194]. For stronger cryptographic protection in high-stakes settings, homomorphic encryption (HE) enables computation over encrypted updates; the CKKS scheme, for example, is widely used for approximate arithmetic on real-valued vectors and has been incorporated into FL toolchains to keep client updates encrypted throughout the aggregation process [195,196]. Taken together, these techniques form a broader privacy-preserving collaboration stack, within which FL has emerged as a visible and increasingly discussed framework in this area. Battery-specific evidence in this direction is beginning to emerge. Recent studies have explored FL for battery diagnostics and prognosis across heterogeneous owners without centralizing raw current-voltage-temperature streams, reporting encouraging results in communication efficiency and cross-site predictive performance [197,198]. For RUL prediction specifically, early FL-based studies suggest that distributed training across multiple fleets or data owners can outperform siloed baselines while preserving data ownership [199,200]. Nevertheless, these results should be interpreted with appropriate caution. In current battery academia, FL is not yet a routine or universally realistic solution, because its practical effectiveness depends on conditions that are often difficult to satisfy simultaneously, including sufficiently large multi-owner cohorts, compatible data standards, manageable communication overhead, and tolerable cross-site heterogeneity in chemistry, protocol, and end-of-life definition. Accordingly, the near-term value of FL is likely to be greater in multi-owner industrial ecosystems or cross-organizational battery-management settings than in routine single-laboratory academic studies. In this sense, privacy-preserving collaborative learning should be regarded as a promising but scenario-dependent future direction for battery health management, rather than as a universally deployable near-term solution.

Uncertainty-aware prognostics and risk-informed decision-making. Most existing RUL models report point estimates and conventional error metrics (RMSE, MAE, R^2), but offer little information about how trustworthy a given prediction is under shifting operating conditions or unobserved failure modes. Recent benchmarks explicitly targeting uncertainty quantification (UQ) in deep prognostics highlight that predictive intervals are often miscalibrated and that different UQ schemes—ensembles, Monte Carlo dropout, Bayesian neural networks (BNNs), or evidential approaches can produce highly divergent risk assessments even at similar point accuracy [201]. For batteries in particular, probabilistic machine learning frameworks argue that understanding epistemic and aleatoric uncertainty is essential to bridge lab-scale RUL models with field deployment and safety assessments [173]. Going forward, RUL research is likely to pivot from "Who fits the trajectory best?" to "Who provides calibrated, decision-useful uncertainty under domain shift?". This includes architectures that natively output predictive distributions, systematically calibrated with proper scoring rules; Bayesian deep learning frameworks that decompose uncertainty sources and propagate them through maintenance cost models [202]. In an operational setting, such models can be coupled with risk-sensitive criteria (e.g., chance constraints, CVaR) so that maintenance thresholds, warranty triggers, and derating policies are optimized not only on expected RUL but also on the tail behavior of predictive distributions [173,201].

Interpretable machine learning for aging-factor discovery and mechanism attribution. Beyond improving predictive accuracy, an increasingly important direction is to use interpretable machine learning to identify the key factors that govern battery aging and to distinguish which parts of the input signal are truly informative for degradation prognosis. In this perspective, explainability is not merely a post hoc tool for model auditing, but a means of connecting statistical prediction with mechanism-oriented understanding. Recent work has shown that interpretable models and explainable feature spaces can be used to organize complex degradation behavior across large protocol spaces, revealing how cycling conditions, trajectory features, and electrochemical state descriptors jointly shape aging outcomes [49]. For future RUL research, this suggests a shift from asking only "which model predicts best" to also asking "which stressors, regimes, and signal patterns most strongly determine lifetime variation." Such a shift could improve scientific insight, feature design, and dataset construction simultaneously. Looking forward, interpretable ML is likely to become increasingly valuable for identifying transferable aging factors across chemistries and operating conditions, testing the stability of feature attribution under distribution shift, and generating mechanism-guided hypotheses that can be further validated experimentally.

Self-supervised and semi-supervised representation learning. Battery fleets already generate massive volumes of current-voltage-temperature time series under diverse drive cycles and ambient conditions, yet only a small fraction of this data is labeled with precise RUL or cycle-to-failure endpoints. This imbalance motivates a shift from fully supervised RUL regression toward representation learning regimes that can exploit unlabeled trajectories. From a methodological perspective, recent surveys on deep-learning-based RUL prediction suggest that self-supervised learning (SSL) and semi-supervised learning (Semi-SL) are natural next steps, particularly for domains where degradation runs are expensive or time-consuming to acquire [203]. In the battery domain, semi-supervised co-estimation frameworks and partial-label methods already demonstrate that unlabeled or partially labeled cycles can meaningfully improve RUL estimation when combined with LSTM- style encoders [204]. More recently, degradation-oriented SSL schemes pretrain encoders on large corpora of raw operating trajectories using predictive, contrastive, or masking objectives tailored to degradation dynamics [204,205]. Fine-tuning these encoders on limited RUL labels tends to improve data efficiency and robustness across chemistries and protocols. For future RUL research, this points toward "foundation" encoders trained once on heterogeneous fleets (possibly under FL or multi-domain setups) and then adapted to specific use cases, rather than repeatedly training task-specific models from scratch.

Large-scale and foundation-style predictive models. Another emerging trend is the rapid development of large-scale predictive models trained across broader operating domains, chemistries, and dataset collections. Most existing battery RUL models are still developed and validated within relatively narrow experimental regimes, which limits their ability to generalize across protocols, temperatures, cell formats, and degradation pathways [206]. Recent studies, however, are beginning to move toward larger and more heterogeneous learning setups, including cross-condition lifetime prediction, time-series foundation models for degradation forecasting, and knowledge-distillation pipelines that transfer broadly pretrained temporal representations into smaller battery-specific predictors. These developments suggest that future battery prognostics may increasingly rely on pretrained representations learned from large multi-source corpora, followed by lightweight adaptation to particular chemistries, operating regimes, or deployment constraints [207,208]. At the same time, this direction raises new challenges in dataset harmonization, label consistency, computational cost, and the balance between generality and physical fidelity. A promising future route is therefore not simply "larger models," but scalable battery learning frameworks that combine broad pretraining, domain adaptation, and deployment-aware compression while preserving robustness under cross-domain shift.

Decision-centric integration with (deep) reinforcement learning. Finally, there is growing recognition that RUL prediction is not an end in itself but an input to maintenance and operation decisions. Instead of evaluating models solely on predictive accuracy, several works integrate RUL estimators into reinforcement learning (RL) or deep RL frameworks that directly optimize long-term reward—availability, safety, and cost—under stochastic degradation [209]. In these settings, the RUL model becomes part of the environment or the agent's belief state, and policies are learned that map health trajectories and uncertainty to actions such as continue, derate, repair, or replace. Survey articles on RL-based predictive maintenance emphasize that such decision-centric formulations can substantially reduce maintenance costs compared with fixed-interval or simple threshold rules, especially when combined with probabilistic RUL outputs and explainability tools (e.g., Shapley analyses of policy drivers) [210]. Early studies also explore multi-agent RL frameworks that coordinate maintenance decisions across fleets or networked assets, with RUL predictors providing a shared notion of residual risk [211]. For RUL research, this suggests a shift from isolated regressors toward tightly coupled "predict-decide-act" pipelines, where model design, uncertainty quantification, and control policy learning are co-optimized under realistic cost and safety constraints.

7. Conclusions

In this review, we combined comprehensive bibliographic tagging with systematic data-driven analysis to provide a rigorous yet accessible overview of machine-learning-based approaches for lithium-ion battery RUL prediction. Beyond summarizing the current state of the art, the quantitative analyses of method usage and temporal trends also enabled us to sketch a reasoned outlook on possible future research trajectories in RUL prognostics.

This article is intended to serve as a reference for both interested beginners and established researchers in this field. On one hand, we offer detailed descriptions of a wide range of methods and provide a careful, data-informed assessment of their usage frequencies, roles, and limitations in battery RUL prediction. Although many algorithms such as LSTM, SVR, CNN, and their variants have been proposed, our synthesis helps readers identify which techniques currently constitute the core toolbox for RUL prediction and which approaches are gradually being phased out. This can help beginners focus on the most influential and practically relevant methods rather than being overwhelmed by the breadth of the literature. On the other hand, the review is also designed as a reference for domain experts. By analyzing the "popularity trends" of mainstream methods over time, we do not merely list which techniques are widely used and currently on the rise, but also discuss why certain methods have gained traction in terms of capability, robustness, and ecosystem support. This perspective allows researchers to align their methodological choices with emerging, high-potential directions and to avoid investing in approaches that the community is moving away from, thereby reducing unproductive detours in research.

Looking forward, the next phase of Li-ion RUL research is unlikely to be driven by a single "best" model family, but rather by a set of complementary directions that collectively address realism, robustness, and deployability. First, hybrid paradigms that fuse sequence learners, margin-based regressors, and physics-informed structure are expected to remain central, as they can better balance reliability, efficiency, generalization, and interpretability than any standalone technique. Second, privacy-preserving collaboration, most notably federated learning, potentially augmented with secure aggregation, DP, and encryption would offer a practical route to scaling model development under real-world constraints. Third, the field is moving beyond point estimates toward uncertainty prognostics, where calibrated predictive distributions and risk-sensitive decision rules are essential under domain shift. What's more, self-supervised and semi-supervised representation learning can further exploit abundant unlabeled operating trajectories, improving data efficiency and adaptation across chemistries and protocols. Ultimately, decision-centric integration with deep reinforcement learning emphasizes that RUL prediction should serve maintenance and operation, encouraging closed-loop pipelines that jointly optimize prediction, uncertainty, and policy under realistic cost and safety constraints.

Overall, this review not only offers a comprehensive account of the current landscape of machine learning methods for battery RUL prediction, but also, through a deliberately objective and data-driven lens, points to several promising directions for future work. We are confident that the perspectives and analyses presented here will provide meaningful guidance, stimulate deeper reflection, and help accelerate progress in this rapidly evolving research area.

Data availability

Data will be made available on request.

CRediT authorship contribution statement

Kaifei Tu: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Wei Yuan:** Writing – review & editing, Funding acquisition. **Xuyang Wu:** Writing – review & editing, Validation. **Xiaoqing Zhang:** Validation, Supervision. **Qing Liu:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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